

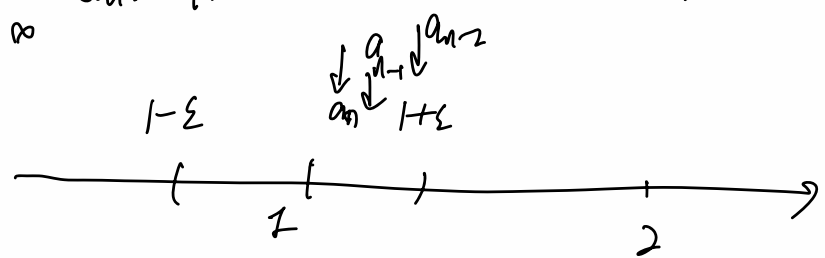
1.4 § 函数的极限

数列的定义: 函数 $f(x)$, 定义域为 $\mathbb{N}$, 值域 $\subseteq \mathbb{R}$.

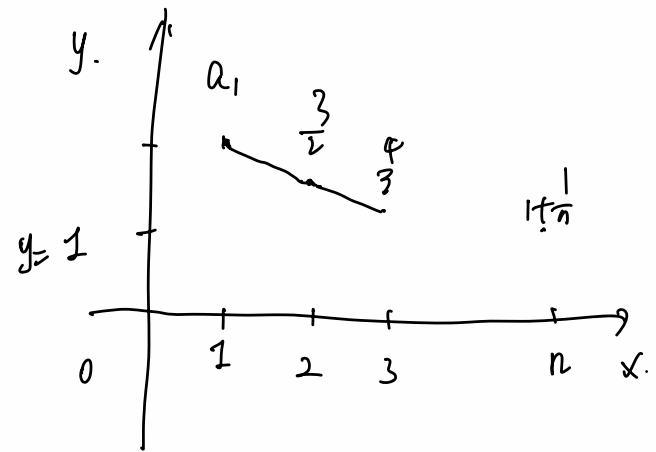
$$\begin{aligned} & \underline{1}, \underline{2}, \dots, \underline{n}, \dots = a_n = f(n) \quad n \in \mathbb{N}. \\ & f(1), f(2), \dots, f(n), \dots \end{aligned}$$

$a_n = 1 + \frac{1}{n}$, $\lim_{n \rightarrow \infty} a_n = 1$. ~~取~~ $\varepsilon > 0$, $N = N_\varepsilon > 0$, $n > N_\varepsilon$ 时 $|a_n - 1| < \varepsilon$

xOy 坐标



$$G = \{ (x, f(x)) \mid x \in D_f \}$$



定义域:

$$(0, +\infty)$$

单调减. 最小下界 1, 一一对应 (可逆)

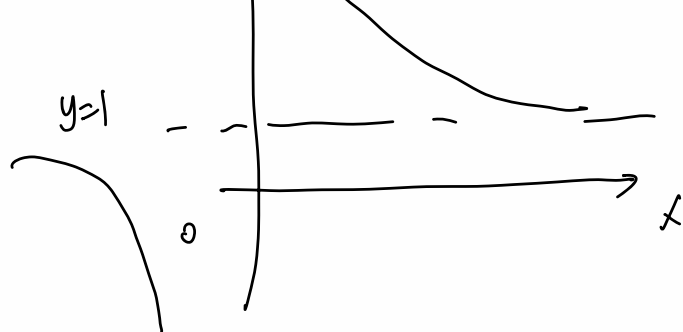
$\forall M > 0, \exists x_0 \in (0, +\infty)$, s.t. $f(x_0) \geq M$.

$$y = 1 + \frac{1}{x}$$



$$x_0 = \frac{1}{M-1}$$

$$x = \frac{1}{y-1} \Leftrightarrow y = 1 + \frac{1}{x}$$



$$\lim_{x \rightarrow +\infty} f(x) = 1$$

(ε - N 语言). $\{a_n\}$. $\lim_{n \rightarrow \infty} a_n = a \Leftrightarrow \forall \varepsilon > 0, \exists N$, s.t. 当 $n > N$ 时
 $|a_n - a| < \varepsilon$ 成立.

无限区间 $[a, +\infty)$, $(-\infty, +\infty)$

(ε - x 语言). 设函数 $f(x)$ 在 $(a, +\infty)$ 上有定义, A 是一个给定的常数.

若 $\forall \varepsilon > 0$, 总存在一个 $X > 0$, s.t. 当 $x > X$ 时
 $|f(x) - A| < \varepsilon$ 成立.

记作 $\lim_{x \rightarrow +\infty} f(x) = A$ 或 $f(x) \rightarrow A$ ($x \rightarrow +\infty$)

$x \rightarrow -\infty$ 函数极限定义

$(-\infty, b]$ $\mathbb{R}$.

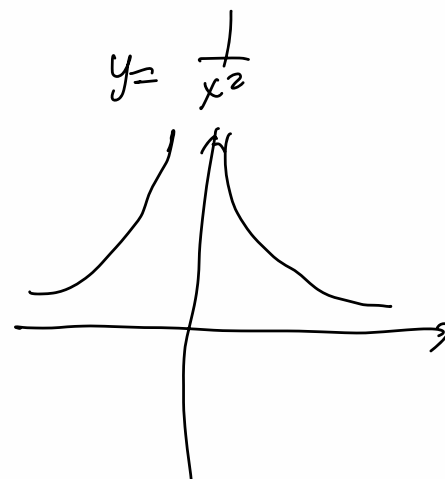
设函数 $f(x)$ 在 $(-\infty, b)$ 有定义, $A \dots$

若 $\forall \varepsilon > 0, \exists X > 0$. s.t. 当 $x < -X$ 时. 总有
 $|f(x) - A| < \varepsilon$ 成立.

记作 $\lim_{x \rightarrow -\infty} f(x) = A$ 或 $f(x) \rightarrow A$ ($x \rightarrow -\infty$)

$a < b$

$\mathbb{R} \setminus \{0\}$.



$x \rightarrow \infty$ 正数极限

设函数 $f(x)$ 在 $(-a, a) \cup (b, +\infty)$ 上有定义, $A \dots$

若 $\forall \varepsilon > 0, \exists X > 0, \text{ s.t. 当 } |x| > X \text{ 时, 有}$

$$|f(x) - A| < \varepsilon \text{ 恒成立.}$$

记为 $\lim_{x \rightarrow \infty} f(x) = A$ 或 $f(x) \rightarrow A (x \rightarrow \infty)$

$$-10^{100} < -(\frac{1}{10})^{100}$$

$$-(\frac{1}{10})^{100}$$

ε $x \rightarrow +\infty$ 任何正数.

$$\{x, f(x) \mid x \in (X, +\infty)\}$$

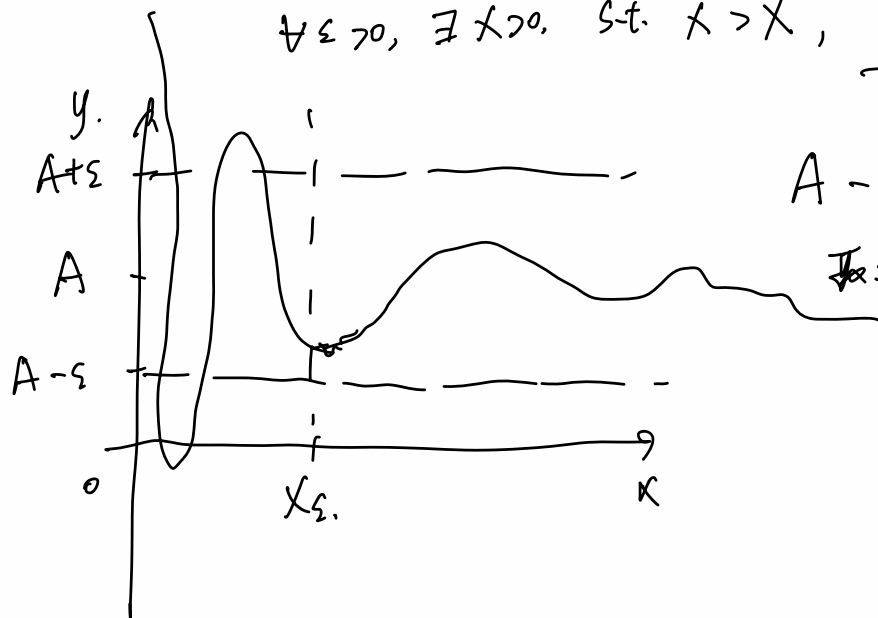
$$\forall \varepsilon > 0, \exists X > 0, \text{ s.t. } x > X, \quad |f(x) - A| < \varepsilon.$$

$\Downarrow$

$$A - \varepsilon < f(x) < A + \varepsilon$$

取 $\varepsilon > 0, X_\varepsilon$

$$y = 1 + \frac{1}{x}$$



例: $\lim_{x \rightarrow \infty} (1 + \frac{1}{x}) = 1.$

解: $\forall \varepsilon > 0, \quad |1 + \frac{1}{x} - 1| < \varepsilon$

$$\Leftrightarrow |\frac{1}{x}| < \varepsilon$$

$$\Leftrightarrow |x| > \frac{1}{\varepsilon}$$

$$\text{取 } X = \frac{1}{\varepsilon}$$

$$\text{取 } x = \frac{1}{\varepsilon} \text{ 且 } |x| > X = \frac{1}{\varepsilon}$$

$$\text{都有 } |f(x) - 1| = \left| \frac{1}{x} \right| < \varepsilon$$

$$\text{例: } \lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2} \quad \lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}.$$

$$\text{命题: } \lim_{x \rightarrow \infty} f(x) = A \text{ 的充必要条件是 } \lim_{x \rightarrow +\infty} f(x) = A \text{ 且 } \lim_{x \rightarrow -\infty} f(x) = A.$$

条件 A, B

$$A \Rightarrow B.$$

充分条件

必要条件

$$A \Leftrightarrow B$$

充要条件

必要性: 结论 $\Rightarrow$ 条件.

充分性: 条件 $\Rightarrow$ 结论.

一. 证明 A 是 B 的充要条件. A 是条件, B 是结论. $A \Rightarrow B$ 充分性, $B \Rightarrow A$ 必要性.

二. 证明 A 的充要条件是 B . B 是条件, A 是结论. $A \Rightarrow B$ 充分性, $B \Rightarrow A$ 必要性.

必要性 左 $\Rightarrow$ 右: $\forall \varepsilon > 0, \exists X > 0, \text{ s.t. } |x| > X, \text{ 有 } |f(x) - A| < \varepsilon.$

取同样的 $X > 0, x > X \Rightarrow \dots$

$x < -X \Rightarrow \dots$

充分性 右 $\Rightarrow$ 左: $\forall \varepsilon > 0, \exists X_1 > 0, \text{ s.t. } \text{当 } x > X_1, \text{ 有 } |f(x) - A| < \varepsilon.$

$$\exists X_2 > 0, \text{ s.t. } \forall x < -X_2 \text{ 时, } |f(x) - A| < \varepsilon.$$

找到 $X > 0$, s.t. $|x| > X$ 时 有 $\dots$

$$\left. \begin{array}{l} \forall x > 0, x > X \geq X_1. \\ \forall x < 0, x < -X \leq -X_2. \end{array} \right\} \Rightarrow \begin{cases} X \geq X_1 \\ X \geq X_2. \end{cases}$$

$$\text{取 } X = \max\{X_1, X_2\},$$

$$X = X_1 + X_2 > 0.$$

例: $\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}$ $\lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}.$

证: $\forall \varepsilon > 0, \quad \left| \arctan x - \frac{\pi}{2} \right| < \varepsilon$

$$\Leftrightarrow \left| \arctan \frac{1}{x} \right| < \varepsilon \quad (x > 0)$$

$$\Leftrightarrow \arctan \frac{1}{x} < \varepsilon \quad \underline{\tan x}$$

$$\Leftrightarrow \frac{1}{x} < \tan \varepsilon$$

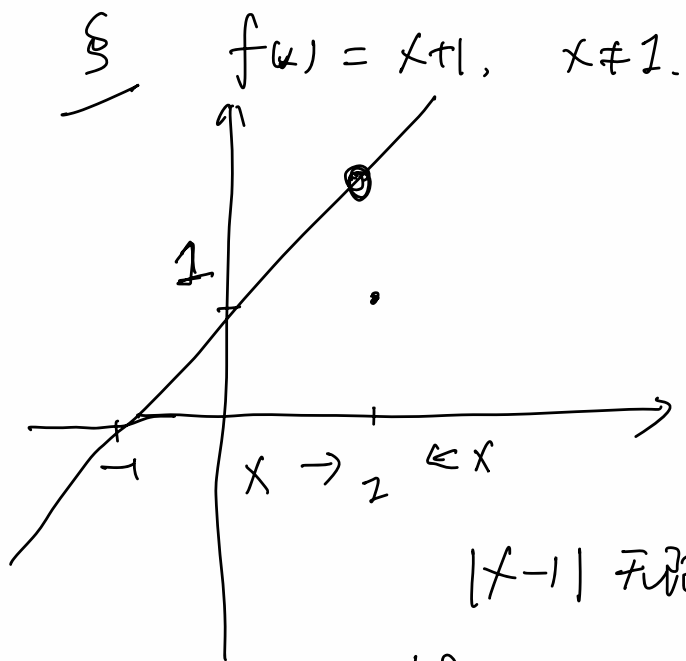
$$\Leftrightarrow x > \frac{1}{\tan \varepsilon}$$

取 $X = \frac{1}{\tan \varepsilon}$ $\forall x > X$ 时,

$$\left| \arctan x - \frac{\pi}{2} \right| < \varepsilon$$

① $\arctan x + \operatorname{arccot} x = \frac{\pi}{2}$

②. $\operatorname{arccot} x = \arctan \frac{1}{x}$
($x > 0$)



$|x-1|$ 无限接近于 0.

$|f(x) - 2|$ 无限接近于 0.

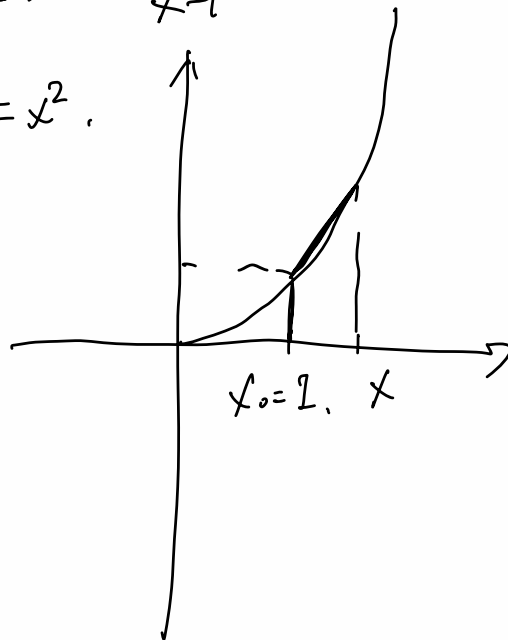
$$|f(x) - 2| < \left(\frac{1}{10}\right)^{100}$$

$$|x - 1| < \left(\frac{1}{10}\right)^{100}$$

$$g(x) = \begin{cases} f(x), & x \neq 1 \\ 1, & x = 1 \end{cases}$$

$$f_1(x) = \frac{x^2-1}{x-1}, x \neq 1$$

$$y = x^2.$$



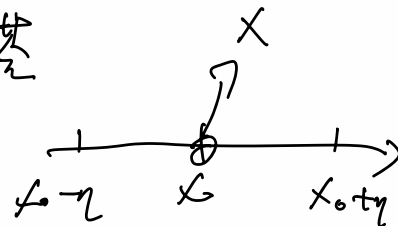
$(1, 2)$

$$\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = ?$$

$$h(x) = x+1, x \in \mathbb{R}.$$

η η η η: 任意正数

$$U_\eta(x_0) = \{0 < |x - x_0| < \eta\}.$$



1.4.2. § 当 $x \rightarrow x_0$ 时的函数极限

(ε-δ 语言)

↓

1.1.4.2

叙述: 当 x 无限接近 x_0 , $f(x)$ 无限接近于 A .
 A 为常数
 若 $\forall \varepsilon > 0, \exists \delta > 0. (\delta < \eta)$
 s.t. 当 $x \in U_\delta(x_0)$, 有 $|f(x) - A| < \varepsilon$ 恒成立.

定义: $\lim_{x \rightarrow x_0} f(x) = A$ 或 $f(x) \rightarrow A (x \rightarrow x_0)$

例: 证 $\lim_{x \rightarrow a} \sin x = \sin a$.

arc sin

$\sin x < x < \tan x$

解: $\forall \varepsilon > 0, \quad \underbrace{|\sin x - \sin a| < \varepsilon.}$

$\Leftrightarrow 2 \left| \cos \frac{x+a}{2} \sin \frac{x-a}{2} \right|$

$\Leftrightarrow 2 \underbrace{\left| \cos \frac{x+a}{2} \right|}_{\leq 1} \left| \sin \frac{x-a}{2} \right| < \varepsilon$

$\Rightarrow 2 \left| \sin \frac{x-a}{2} \right| < \varepsilon.$

$\Rightarrow 2 \cdot \frac{|x-a|}{2} < \varepsilon$

$\Leftrightarrow |x-a| < \varepsilon.$

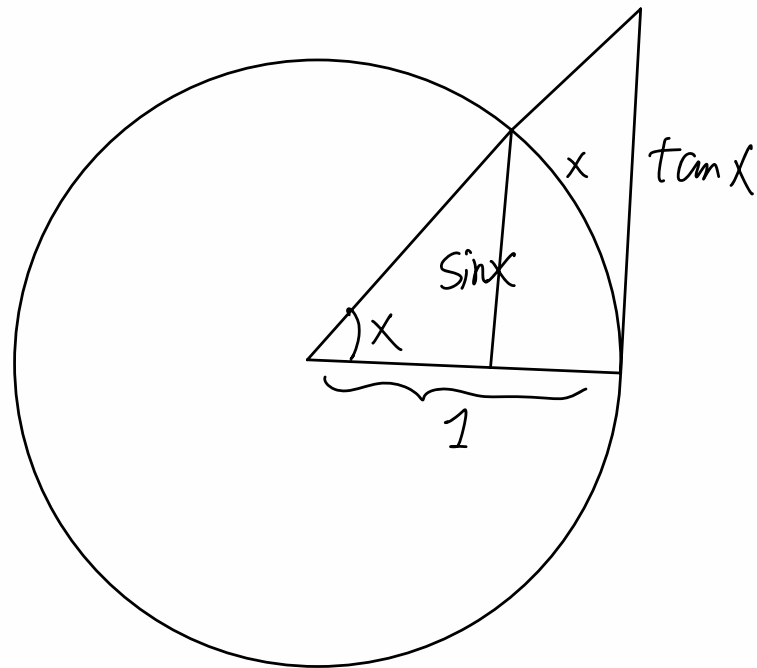
取 $\delta = \varepsilon$. 那么当 $|x-a| < \delta$ 时, 有

$|\sin x - \sin a| \leq |x-a| < \delta = \varepsilon.$

例: 证明当 $x_0 > 0, \quad \lim_{x \rightarrow x_0} \sqrt{x} = \sqrt{x_0}.$

$a^2 - b^2 = (a+b)(a-b).$

解: $\forall \varepsilon > 0, \quad \underbrace{|\sqrt{x} - \sqrt{x_0}| < \varepsilon.} \quad \frac{\sqrt{x} - \sqrt{x_0}}{1} \cdot \left(\frac{\sqrt{x} + \sqrt{x_0}}{\sqrt{x} + \sqrt{x_0}} \right) \left\{ \begin{array}{l} a+b \\ a-b \end{array} \right.$ 其平方因式



$$\Leftrightarrow \left| \frac{1}{\sqrt{x} + \sqrt{x_0}} \right| < \frac{1}{\sqrt{x_0}} \Rightarrow \frac{1}{\sqrt{x_0}} \cdot |x - x_0| < \varepsilon.$$

$$\forall x > 0, \quad \sqrt{x} > 0. \quad \Rightarrow |x - x_0| < \sqrt{x_0} \varepsilon.$$

$$\sqrt{x} + \sqrt{x_0} > \sqrt{x_0}$$

$$\frac{1}{\sqrt{x} + \sqrt{x_0}} < \frac{1}{\sqrt{x_0}}$$

$$\text{取 } \delta = \sqrt{x_0} \varepsilon$$

$$0 < |x - x_0| < \delta \Rightarrow \{x > 0\}.$$

$$\text{且 } x \in \{x > 0\}.$$

$$0 < x_0 - \delta < x$$

$$\delta = \min \{ \varepsilon \sqrt{x_0}, x_0 \}.$$

$$\text{例: } \lim_{x \rightarrow 2} x^2 = 4.$$

$$\text{证: } \forall \varepsilon > 0, \quad |x^2 - 4| = |(x-2)(x+2)| < \varepsilon$$

$$0 < |x-2| < 1 \Rightarrow 1 < x < 3$$

$$|x+2| < 5.$$

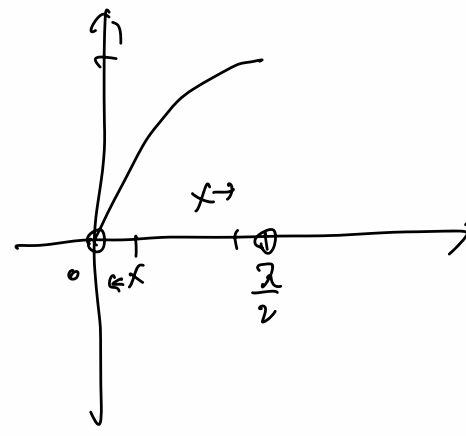
$$\Rightarrow 5|x-2| < \varepsilon.$$

$$\Leftrightarrow |x-2| < \frac{\varepsilon}{5}.$$

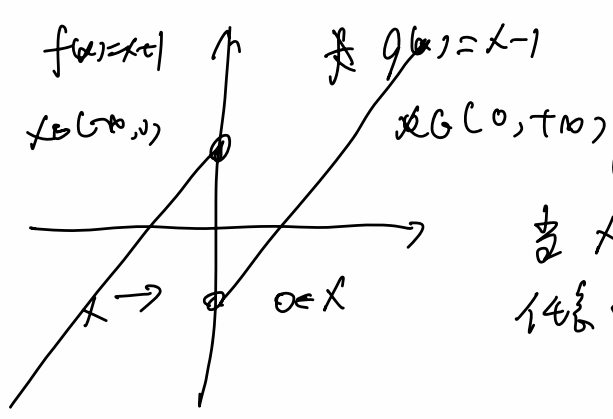
$$\text{取 } \delta = \min \left\{ \frac{\varepsilon}{5}, 1 \right\} \text{ 则 } \begin{cases} |x-2| < \delta \\ |x-2| < 1 \end{cases}$$

$$\S. \quad f(x) = \sin x \quad x \in \left(0, \frac{\pi}{2}\right)$$

$\varepsilon-\delta$ 叙述极限存在.



左极限. - - - 不存在



$(\varepsilon-\delta)$ 语言.

当 x 无限接近于 0 时.
任意给定的 $A > 0$ 都不满足

§ 左右极限.

右极限 $(\varepsilon-\delta)$. 设函数 $f(x)$ 在 $(x_0, x_0 + \eta)$ 有定义. A 是常数.

$\forall \varepsilon > 0, \exists \delta, \text{ s.t. } x \in (x_0, x_0 + \delta) \text{ 时 } |f(x) - A| < \varepsilon \text{ 成立}$

$\lim_{x \rightarrow x_0^+} f(x) = A$ 或 $f(x) \rightarrow A \ (x \rightarrow x_0^+)$.

左极限 - - - 在 $(x_0 - \eta, x_0)$ 有定义.

$\forall \varepsilon > 0, \exists \delta, \text{ s.t. } x \in (x_0 - \delta, x_0) \text{ 时 } |f(x) - A| < \varepsilon \text{ 成立}.$

记为 $\lim_{x \rightarrow x_0^-} f(x) = A$ 或 $f(x) \rightarrow A \ (x \rightarrow x_0^-)$

$x \rightarrow \infty \Leftrightarrow \begin{matrix} x \rightarrow +\infty \\ x \rightarrow -\infty \end{matrix}$ $|x| > \underline{X}$ $\underline{x_1, x_2}$ $\forall \varepsilon > 0, \begin{matrix} \delta_1 \\ \delta_2 \end{matrix}$

命题: $\lim_{x \rightarrow x_0} f(x) = A$ 的充要条件是: $\lim_{x \rightarrow x_0^+} f(x) = A$ 且 $\lim_{x \rightarrow x_0^-} f(x) = A$.
左右极限不相等.

例: $\lim_{x \rightarrow 0} \frac{|x|}{x}$ 的极限不存在 $\text{sgn } x$
定义域 $x \neq 0$.

解: 当 $x > 0$ 时, $\frac{|x|}{x} = \frac{x}{x} = 1$. $y = C$.
 $\forall x_0 \in D_f = \mathbb{R}$.

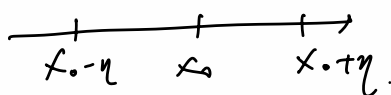
$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1 \quad (\varepsilon - \delta \text{ 证 } \frac{1}{2})$$

$$\lim_{x \rightarrow x_0} f(x) = C.$$

$$\text{当 } x < 0 \quad \frac{|x|}{x} = \frac{-x}{x} = -1.$$

↓ ↓

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1.$$



$$\lim_{n \rightarrow \infty} x_n \rightarrow x_0.$$

$$\text{当 } n > N, \quad x_n \in D_f$$

Heine 定理

设函数 $f(x)$ 在 $U_p(x_0)$ 有定义.

$$\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \{x_n\} \quad x_n \rightarrow x_0, (x_n \neq x_0).$$

$$\text{都有 } \lim_{n \rightarrow \infty} f(x_n) = A.$$

证明: 充分性: 左 $\Rightarrow$ 右.

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } 0 < |x - x_0| < \delta \text{ 有 } |f(x) - A| < \varepsilon \text{ 恒成立.}$$

$$\forall \varepsilon > 0, \exists N > 0,$$

$$\text{s.t. } n > N \text{ 时}$$

$$|x_n - x_0| < \varepsilon'$$

$$\text{得到 } |f(x_n) - A| < \varepsilon.$$

$$\{x_n \neq x_0\}.$$

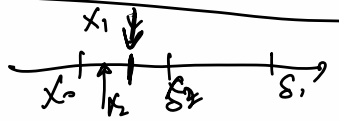
$\forall \varepsilon > 0, \exists N$ 使得 $n > N$ 时 $|x_n - x_0| < \varepsilon$.
 得到 δ , $\rightarrow$ $|f(x_n) - A| < \varepsilon$.

充分性: 右 $\Rightarrow$ 左.

反证法. 假设左不成立. 若 $\lim_{x \rightarrow x_0} f(x) \neq A$.

$\exists \varepsilon_0 > 0$, s.t. $\forall \delta > 0$, $\exists y \in \overset{\circ}{U}_\delta(x_0)$, $|f(y) - A| \geq \varepsilon_0$ 成立.

取 $\delta = \frac{1}{n}$, $n \in \mathbb{N}$



取 $\delta_1 = 1$, $\exists x_1$, $0 < |x_1 - x_0| < \delta_1$, s.t. $|f(x_1) - A| \geq \varepsilon_0$.

取 $\delta_2 = \frac{1}{2}$, $\exists x_2$, $0 < |x_2 - x_0| < \frac{1}{2} \leq \delta_2$, s.t. $|f(x_2) - A| \geq \varepsilon_0$.
 $0 < |x_2 - x_0| < |x_1 - x_0|$

$\vdots$
 取 $\delta_3 = \frac{1}{3}$, $\exists x_3$. $0 < |x_3 - x_0| < |x_2 - x_0| < |x_1 - x_0|$

$\vdots$
 取 $\delta_n = \frac{1}{n}$, $\exists x_n$,

得到数列 $\{x_n\}$ $x_n \neq x_0, \forall n \in \mathbb{N}$.

用 $(\varepsilon - N)$ 证 $\lim_{n \rightarrow \infty} x_n = x_0$.

$\forall \varepsilon > 0$. s.t. $\left(\frac{1}{n \varepsilon}\right) < \varepsilon < \frac{1}{n}$

$|f(x_n) - A| \geq \varepsilon_0$.

$\lim_{n \rightarrow \infty} f(x_n) \neq A$.

$\frac{2}{\sqrt{b}}$

方法一: 极限不存在.

二: 已知函数极限存在 $\Rightarrow$ 数列极限存在.

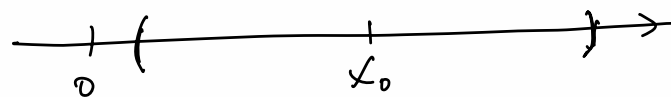
$$\left\{ \begin{array}{l} \lim_{x \rightarrow 0} \sin x = \sin 0 = 0. \\ \lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0. \end{array} \right. \quad x_n = \frac{1}{n^2}, \quad \forall n \in \mathbb{N}, \frac{1}{n^2} \neq 0.$$

例 1.4.2. 证明 当 $x_0 > 0$ 时, $\lim_{x \rightarrow x_0} \sqrt{x} = \sqrt{x_0}$.

证明: 首先注意到: $\sqrt{x}$ 的定义域为 $x \geq 0$.

$$\forall \varepsilon > 0. \quad \underbrace{|\sqrt{x} - \sqrt{x_0}| < \varepsilon}_{\downarrow} \quad \underbrace{x \in \bigcup_{\sqrt{x_0} \pm \varepsilon} (x_0)}_{\substack{x_0 - \delta \quad x_0 + \delta}} \subseteq \underbrace{\{x \geq 0\}}$$

利用共轭因式 $\frac{|x - x_0|}{|\sqrt{x} + \sqrt{x_0}|} \leq \frac{|x - x_0|}{\sqrt{x_0}} < \varepsilon$



$x \geq 0$ 且 $0 < |x - x_0| < \sqrt{x_0} \varepsilon$ 直接取 $\delta = \sqrt{x_0} \varepsilon$ 可不可以? ① 若 $\sqrt{x_0} \varepsilon \leq x_0$, 则取 $\delta = \sqrt{x_0} \varepsilon$
② 若 $\sqrt{x_0} \varepsilon > x_0$, 则取 $\delta = x_0$.
} $\delta = \min\{x_0, \sqrt{x_0} \varepsilon\}$
 $\delta = \frac{1}{2} x_0$

Heine 定理: 设函数 $f(x)$ 在 x_0 的某去心邻域内有定义, 则

$$\lim_{x \rightarrow x_0} f(x) = A \iff \text{对于任何一个收敛到 } x_0 \text{ 的数列 } \{x_n\}, x_n \neq x_0, \\ \text{有 } \lim_{n \rightarrow \infty} f(x_n) = A.$$

$x_1, \dots, x_N, \dots$

证明: 充分性 $\Leftarrow$ 不妨设 $\{x_n\} \subseteq x_0$ 的去心邻域中.

条件: 右边成立

用反证法, 假设右边不成立,

$$\lim_{x \rightarrow x_0} f(x) \neq A$$

$\exists \varepsilon_0 > 0$, s.t. $\forall \delta > 0$, 都 $\exists x \in U_\delta(x_0)$, s.t. $|f(x) - A| \geq \varepsilon_0$.

取 $\delta_1 = 1$, $\exists x_1 \in \{0 < |x - x_0| < 1\}$, s.t. $|f(x_1) - A| \geq \varepsilon_0$.

取 $\delta_2 = \frac{1}{2}$, $\exists x_2 \in \{0 < |x - x_0| < \frac{1}{2}\}$, s.t. $|f(x_2) - A| \geq \varepsilon_0$.

取 $\delta_3 = \frac{1}{3}$, $\exists x_3 \in \dots \frac{1}{3} \dots$

$\vdots$

取 $\delta_n = \frac{1}{n}$, $\exists x_n \in \{0 < |x - x_0| < \frac{1}{n}\}$, s.t. $|f(x_n) - A| \geq \varepsilon_0$.

$\vdots$

构造了数列 $\{x_n\}$.

要说明 $\lim_{n \rightarrow \infty} x_n = x_0$.

$\forall \varepsilon > 0, \exists N_\varepsilon > 0$, s.t. $\forall n > N_\varepsilon$ 时有 $|x_n - x_0| < \varepsilon$.

$\{\frac{1}{n}\}$

$\forall \varepsilon > 0$, 总能找到一个正整数 m , s.t. $0 < \frac{1}{m} < \varepsilon$.

$\frac{1}{\varepsilon} > 0$. $[\frac{1}{\varepsilon}] \leq \frac{1}{\varepsilon} < [\frac{1}{\varepsilon}] + 1$. 令 $\underline{M} = [\frac{1}{\varepsilon}] + 1$ $\frac{1}{M-1} \leq \frac{1}{\varepsilon} < \frac{1}{M}$. $\frac{1}{m} < \varepsilon \leq \frac{1}{m+1}$

$n > m$, 有 $|x_n - x_0| < \frac{1}{m} < \varepsilon$.

取 $N_\varepsilon = m$. $\Rightarrow \lim_{n \rightarrow \infty} x_n = x_0$.

例: 证明 $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ 不存在.

① 存在 $x_n \rightarrow 0$, s.t. $\{f(x_n)\}$ 发散

② 找两 $x_n \rightarrow 0, y_n \rightarrow 0$ s.t. $\lim_{n \rightarrow \infty} f(x_n) \rightarrow A$
 $\lim_{n \rightarrow \infty} f(y_n) \rightarrow B$.

证明: $x_n = \frac{1}{2n\pi}$, $n \in \mathbb{N}$, $x_n \neq 0$, $\lim_{n \rightarrow \infty} x_n = 0$.

$\searrow$
 $\sin \frac{1}{x_n} = 0$. $\lim_{n \rightarrow \infty} f(x_n) = 0$.

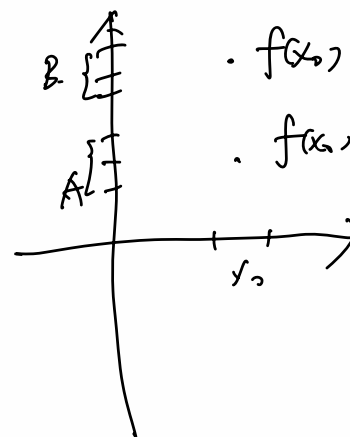
$y_n = \frac{1}{(2n+1)\pi}$ $n \in \mathbb{N}$, $y_n \neq 0$. $\lim_{n \rightarrow \infty} y_n = 0$.

$$\sin \frac{1}{y_n} = 1, \quad \lim_{n \rightarrow \infty} f(y_n) = 1.$$

性质一. (唯一性) 若 $\lim_{x \rightarrow x_0} f(x) = A$ 且 $\lim_{x \rightarrow x_0} f(x) = B$, $A = B$.

(if $A < B$)

2.



性质二 (局部有界性). 有界性: $\exists M > 0$, s.t. $\forall x \in D_f$, 都有 $|f(x)| \leq M$.

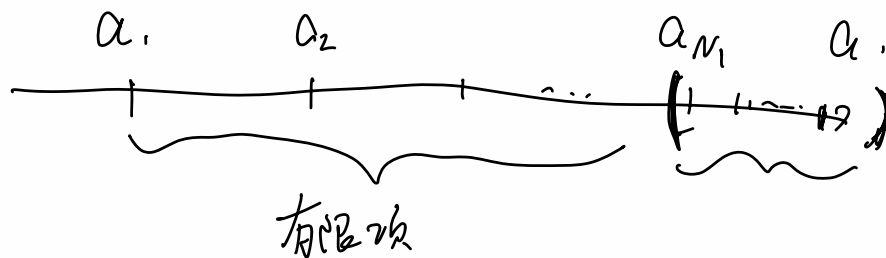
局部有界性: $\exists M > 0$, 和 $\delta > 0$, s.t. $\forall x \in \{x \mid |x - x_0| < \delta\}$.

都有 $|f(x)| \leq M$, 称函数 $f(x)$ 在 x_0 处局部有界.

取 $\varepsilon = 1$, $\delta_1 > 0$, $\forall x \in \overset{\circ}{U}_{\delta_1}(x_0)$, $|f(x)| \leq A + 1$.

$y = x$ 或 $y = \frac{1}{x}$,

取 $\varepsilon = 1$, $N_1 > 0$. 当 $n > N_1$.



let $f = (f_n)$ s.t. $\lim_{n \rightarrow \infty} f_n = A$ $\forall n \in (0, A)$.

性质 = (保号性): 若 $\lim_{x \rightarrow x_0} f(x) = A > 0$,

$\exists \delta > 0$, s.t. $x \in \dot{U}_\delta(x_0)$ 有 $f(x) > \eta > 0$.

$$A < 0. \quad \eta \in (A, 0).$$

$$\dots \dots \dots f(x) < \eta < 0.$$

推论: 在 x_0 的去心邻域内, $\underbrace{f(x) \geq 0}_{f(x) > 0}$, $\lim_{x \rightarrow x_0} f(x) = A$, 则 $\underbrace{A \geq 0}_{\text{不能换}}$.

$$f(x) = |x|. \quad x \rightarrow 0.$$

推论 = : $\lim_{x \rightarrow x_0} f(x) = A \neq 0. \quad \exists \delta > 0$, s.t. $\exists 0 < |x - x_0| < \delta$ 内有

$$\underbrace{\left| \frac{1}{f(x)} \right| < \frac{2}{|A|}}_{|f(x)| > \frac{|A|}{2}}$$

推广. $\left\{ \begin{array}{l} x \rightarrow \infty \\ x \rightarrow x_0, x \rightarrow +\infty, x \rightarrow -\infty \\ x \rightarrow x_0, x \rightarrow x_0+, x \rightarrow x_0- \end{array} \right.$ $\underline{\underline{x \rightarrow \textcircled{x}}}$

... .. 且其极限

$$f(x) \rightarrow A, \quad A \neq \infty$$

定义: 如果 $\lim_{x \rightarrow \odot} f(x) = 0$, 则称函数 $f(x)$ 当 $x \rightarrow \odot$ 时为无穷小量, 记为

$$\underline{\underline{f(x) = o(1) \quad (x \rightarrow \odot)}}$$

例: $x \rightarrow 0, \quad x, \sin x$

$x \rightarrow \infty, \quad \frac{1}{x}, \frac{\sin x}{x}$

$x \rightarrow -\infty, \quad e^x$

$n \rightarrow \infty \quad \left\{ \frac{1}{2^n} \right\}$

注: ①. 无穷小量, 是变量/函数, 且极限过程中.

$$f(x) = \frac{1}{x} \quad (x \rightarrow \infty)$$

②. 非数,

③. 0 常值函数

定理: $\lim_{x \rightarrow \odot} f(x) = A \iff f(x) = A + o(1) \quad (x \rightarrow \odot)$

结论: ①. $\lim_{x \rightarrow \odot} f(x) = A \Rightarrow \lim_{x \rightarrow \odot} |f(x)| = |A|$ 反之不成立.

②. 无穷小量的绝对值也是无穷小量 ($\Leftrightarrow$)

③. $\lim_{x \rightarrow \infty} f(x) = A \Leftrightarrow |f(x) - A| = o(1) \quad (x \rightarrow \infty)$

例: $\operatorname{sgn} x = \begin{cases} 1 & x > 0. \\ 0 & x = 0. \\ -1 & x < 0. \end{cases} \quad |\operatorname{sgn} x| = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}$

$\lim_{x \rightarrow 0} \operatorname{sgn} x$ 不存在 $\quad \lim_{x \rightarrow 0} |\operatorname{sgn} x| = 1.$

定理: 在同一极限过程中, 有限个无穷小可以进行加、减、乘法运算.

证: 只需证明两个无穷小即可,

$(x \rightarrow \infty) \quad \underbrace{a_1 + a_2 + \dots + a_k}_{b_1} \rightarrow a_k$ 结合律.

不妨设 $x \rightarrow a, \quad \lim_{x \rightarrow a} \alpha(x) = 0, \quad \lim_{x \rightarrow a} \beta(x) = 0$

$\forall \varepsilon > 0, \quad \exists \delta_1 > 0, \quad \text{s.t.} \quad 0 < |x - a| < \delta_1 \text{ 时, } |\alpha(x)| < \varepsilon, \quad ①$

$\exists \delta_2 > 0, \quad \dots \dots \dots < \delta_2 \text{ 时, } |\beta(x)| < \varepsilon. \quad ②.$

取 $\delta = \min \{ \delta_1, \delta_2 \}, \quad \text{那么} \quad 0 < |x - a| < \delta.$

$$\left. \begin{aligned} |\alpha(x) \pm \beta(x)| &\leq |\alpha(x)| + |\beta(x)| < 2\varepsilon \\ |\alpha(x)\beta(x)| &\leq |\alpha(x)| |\beta(x)| < \varepsilon^2 \end{aligned} \right\}$$

$$\lim_{x \rightarrow a} [\alpha(x) \pm \beta(x)] = \lim_{x \rightarrow a} [\alpha(x), \beta(x)] = 0.$$

例: $\lim_{n \rightarrow \infty} \underbrace{\frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n}{n^2}}_{A(n)}$

解:

$$\begin{aligned} A(n) &= \frac{1}{n^2} (1 + \dots + n) \\ &= \frac{1}{2} \frac{n+1}{n} = \frac{1}{2} + \frac{1}{2n} \end{aligned}$$

在 $n \rightarrow \infty$ 时 $\frac{1}{n^2}, \dots, \frac{n}{n^2}$
 $\frac{1}{2n}$ 也是无穷小.

$$\lim_{n \rightarrow \infty} A(n) = \frac{1}{2}$$

定理: 局部有界的正数和无穷小的积也是无穷小.

$\exists M > 0, \delta > 0,$

7种.

$x \rightarrow x_0$

$x \rightarrow x_0 +$

s.t. $x \in (x_0, x_0 + \delta)$

$$\underline{x \rightarrow x_0 -} \quad \text{有 } |f(x)| \leq M.$$

$$x \in (x_0 - \delta, x_0)$$

$$\left\{ \begin{array}{ll} x \rightarrow \infty & \{x > |x|\} = (-\infty, -x) \cup (x, +\infty) \\ x \rightarrow +\infty & (x, +\infty) \\ x \rightarrow -\infty & (-\infty, -x) \end{array} \right.$$

证明: 设 $x \rightarrow a$. $\lim_{x \rightarrow a} \alpha(x) = 0$ $f(x)$ 局部有界.

$$\exists M > 0, \delta > 0, \text{ s.t. } 0 < |x - a| < \delta \text{ 有 } |f(x)| < M, \textcircled{1}$$

$\forall \varepsilon > 0$, 想证明 $|\alpha(x) f(x)| < \varepsilon$.

$$|\alpha(x)| |f(x)| < \varepsilon.$$

$$\text{只需 } |\alpha(x)| < \frac{\varepsilon}{M}$$

$$\exists \delta' > 0, \text{ s.t. } \text{当 } 0 < |x - a| < \delta' \text{ 时, 有 } |\alpha(x)| < \frac{\varepsilon}{M} \textcircled{2}.$$

$$\text{取 } \delta_1 = \{ \delta, \delta' \} \quad \text{那么在 } 0 < |x - a| < \delta_1 \text{ 内}$$

$$\Rightarrow |\alpha(x) f(x)| < \varepsilon.$$

$$\lim_{x \rightarrow a} [f(x) \alpha(x)] = 0$$

推论一: 若 A 为常数, $\lim_{x \rightarrow \infty} \alpha(x) = 0$, 则 $\lim_{x \rightarrow \infty} [A \alpha(x)] = 0$.

推论二: 若 $\lim_{x \rightarrow \infty} f(x) = A$, $\lim_{x \rightarrow \infty} \alpha(x) = 0$, 则 $\lim_{x \rightarrow \infty} [f(x) \alpha(x)] = 0$

$f(x)$ 是局部有界的 $\begin{cases} x \rightarrow a & 0 < |x-a| < \delta. \\ x \rightarrow \infty & (-\infty, -X) \cup (X, +\infty) \end{cases}$

推论三: 若 $\lim_{x \rightarrow \infty} f(x) = A \neq 0$, $\lim_{x \rightarrow \infty} \alpha(x) = 0$, 则 $\lim_{x \rightarrow \infty} \frac{\alpha(x)}{f(x)} = 0$.

$|\frac{1}{f(x)}| \leq \frac{2}{|A|}$ 局部有界函数.

例: $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$. $\frac{1}{x} = o(1) (x \rightarrow \infty)$

$|\sin x| \leq 1$.

$\lim_{x \rightarrow \infty} \sin x$ 不存在

$\lim_{x \rightarrow \infty} \frac{\arctan x}{x} = 0$.

$\lim_{x \rightarrow \infty} \arctan x$ 不存在
 $x \rightarrow +\infty \quad \frac{\pi}{2}$
 $x \rightarrow -\infty \quad -\frac{\pi}{2}$

$(-\infty, +\infty)$

$(-\frac{\pi}{2}, \frac{\pi}{2})$

无穷大.

$\{\sqrt{n}\}, 1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \dots, \sqrt{n}, \dots$

给定一个 $M > 0$, 只要 n 充分大, $\sqrt{n} > M$.

总存在 N , s.t. 当 $n > N$ 时 必有 $\sqrt{n} > M$.

$$\sqrt{n} > M \Rightarrow n > M^2 \quad \text{取 } N = [M^2] + 1$$

数列的无穷大:

定义: $\{x_n\}$, 若对 $\forall M > 0$, $\exists N$ s.t. 当 $n > N$ 时, 有 $|x_n| > M$ 成立, 则称 $\{x_n\}$ 是一个无穷大数列.

记为 $\lim_{n \rightarrow \infty} x_n = \infty$ 或 $x_n \rightarrow \infty$ ($n \rightarrow \infty$),

$|x_n| > M$ 改成 $x_n > M$. 正无穷大数列 $\lim_{n \rightarrow \infty} x_n = +\infty$.

改成 $x_n < -M$, 负无穷大数列 $\lim_{n \rightarrow \infty} x_n = -\infty$.

例: 设 $|q| > 1$, $\{q^n\}$ 是无穷大量.

$$\forall M > 0, \quad |q^n| > M$$

$$\Leftrightarrow |q|^n > M.$$

$$\Leftrightarrow n \log |q| > \log M \quad \text{由于 } |q| > 1$$

$$\Leftrightarrow n > \frac{\log M}{\log |q|}$$

$$\text{取 } N = \left\lceil \frac{\log M}{\log |q|} \right\rceil + 1 \quad \text{当 } n > N \text{ 时 } \Rightarrow |q^n| > M.$$

$x \rightarrow x_0$.

定义: 设函数 $f(x)$ 在 x_0 的某个空心邻域内有定义, 如果 $\forall M > 0$, $\exists \delta > 0$,

s.t. $0 < |x - x_0| < \delta$ 时, 都有 $|f(x)| > M$, 则称函数 $f(x)$ 是在 $x \rightarrow x_0$ 时无穷大量,
记 $x \rightarrow x_0$ $f(x) = \infty$.

定义: $x \rightarrow \infty$. 在 $|x| > X'$ 有定义, 如果 $\forall M > 0, \exists X > 0$, s.t. 当 $|x| > X$,
都有 $|f(x)| > M$, 则称 $f(x)$ 当 $x \rightarrow \infty$ 是无穷大量,

记为 $\lim_{x \rightarrow \infty} f(x) = \infty$

在两个定义中, $|f(x)| > M$

$f(x) > M$.

$\lim_{x \rightarrow x_0} f(x) = +\infty$

$\lim_{x \rightarrow \infty} f(x) = +\infty$.

$f(x) < -M$.

$\lim_{x \rightarrow x_0} f(x) = -\infty$

$\lim_{x \rightarrow \infty} f(x) = -\infty$

$\left\{ \begin{array}{l} \lim_{x \rightarrow x_0^+} f(x) = \infty \quad +\infty \quad -\infty \\ \lim_{x \rightarrow x_0^-} f(x) = \infty \\ \lim_{x \rightarrow +\infty} f(x) = \infty \\ \lim_{x \rightarrow -\infty} f(x) = \infty \end{array} \right.$

注: ① 无穷大量是发散的特例

② 无穷大量是变量, ($x \rightarrow \infty$)

③ 函数为无界大 $\Rightarrow$ 无穷函数 (反之不成立)

例: $\lim_{x \rightarrow +\infty} e^x = +\infty$.

证: $\forall M > 0, \quad \underline{\underline{e^x > M}}$

不妨设 $M > 1$.

$$e^x > M$$

$(x, +\infty)$

$$x > \ln M > 0.$$

取 $X = \ln M$, 当 $x > X$ 时 都有 $e^x > M$ 成立.

$$\lim_{x \rightarrow -\infty} e^x = 0.$$

$\lim_{x \rightarrow +\infty} e^x$ 极限不存在

$$\lim_{x \rightarrow +\infty} \ln x = +\infty$$

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$\lim_{x \rightarrow +\infty} x = \infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

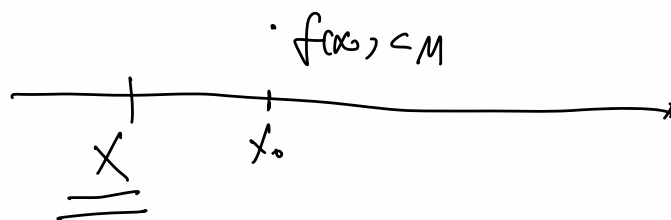
$$x \rightarrow 0^-$$

无界函数: $\forall M > 0, \exists x_0 \in D_f, \text{ s.t. } |f(x_0)| > M.$

一个极限过程不是无穷大:

设 $x \rightarrow +\infty$

$\exists M > 0, \text{ s.t. } \forall X > 0, \text{ 都 } \exists x_0 \in (X, +\infty), \text{ s.t. } |f(x_0)| < M.$



例: 函数 $f(x) = x \cos x, x \in (-\infty, +\infty)$ 非无穷大但无界.

$$\cos(2n\pi) = 1, n \in \mathbb{Z}.$$

$$f(2n\pi) = 2n\pi, n \in \mathbb{Z}.$$

设 $n \in \mathbb{N}, n \uparrow, f(2n\pi) \rightarrow +\infty$ 当 $n \rightarrow +\infty$ 时.

$$\forall M > 0, \exists m \text{ s.t. } M < 2m\pi. f(2m\pi) = 2m\pi > M.$$

$$\text{不妨取 } M = 1, \forall X > 0, \exists m \text{ s.t. } X < 2m\pi + \frac{\pi}{2}.$$

$$f(2m\pi + \frac{\pi}{2}) = 0 < M$$

当 $x \rightarrow +\infty$ 时 $f(x)$ 不是无穷大量.

无穷大数 $\Rightarrow$ 无界 (反之不成立)

1, 0, 2, 0, 3, 0, ..., n, 0, ...
 $\downarrow \quad \downarrow \quad \downarrow \quad \quad \quad \downarrow$

命题: 无界数列一定可以取出无穷大子列.

证: 设 $\{a_n\}$ 是无界数列.

$\forall M > 0, \exists N, \text{ s.t. } |a_N| > M.$

取 $M_1 = 1, \exists N_1, \text{ s.t. } |a_{N_1}| > 1.$

取 $M_2 = 2, \underline{\exists N_2}, \text{ s.t. } |a_{N_2}| > 2.$

先取 $N_2 > N_1$, 若 $\forall n > N_1$ 都有 $|a_n| \leq 2$.

$\vdots$
 $\vdots$
 $\vdots$
 $\underbrace{\quad \quad \quad N_1 - 1 \quad N_1 \quad \quad \quad}_{\text{有限项}} \quad \underbrace{\quad \quad \quad}_{\text{有界}}$

取 $M_3 = 3, \exists N_3 > N_2 > N_1, \text{ s.t. } |a_{N_3}| > 3.$

$\vdots$

取 $M_n = n, \exists N_n > N_{n-1} > \dots > N_1, \text{ s.t. } |a_{N_n}| > n.$

$\vdots$

$\{a_{N_n}\}.$

$a_{N_1}, a_{N_2}, \dots, a_{N_n}, \dots$

$N_1 < N_2 < \dots < N_n < \dots$

证明无穷大

无穷大的运算

① 有限个无限大的乘积是无穷大.

②. $\lim_{x \rightarrow \infty} f(x) = A \neq 0$, $\lim_{x \rightarrow \infty} g(x) = \infty$, 则 $\lim_{x \rightarrow \infty} f(x)g(x) = \infty$.

③. 局部有下界: $|f(x)| > A > 0$, $f(x)$ 与无穷大的积是无穷大

④. 两个正(负)无穷大的和是正(负)无穷大

例: $\lim_{x \rightarrow \infty} x^n = \lim_{x \rightarrow \infty} \underbrace{x \cdots x}_{n \text{ 个}} = \infty$.

$$\lim_{x \rightarrow \infty} x = \infty.$$

n 为奇数 $\lim_{x \rightarrow +\infty} x^n = +\infty$ $\lim_{x \rightarrow -\infty} x^n = -\infty$.

n 为偶数 $\lim_{x \rightarrow \pm\infty} x^n = +\infty$.

$\underbrace{a_n}_{\neq 0}$ $\underbrace{a_{n-1}}$

例: 设 $P_n(x) = \underbrace{a_n x^n}_{\neq 0} + \underbrace{a_{n-1} x^{n-1}} + \cdots + \underbrace{a_1 x + a_0}$ 为实系数多项式. $a_n \neq 0$.

$$\lim_{x \rightarrow \infty} P_n(x) = \infty.$$

证: $\underbrace{x^n, x^{n-1}, \dots, x}_{a_n, a_{n-1}, \dots, a_1}$ 是无穷大.

$x \neq 0$.
 $|x| > 1$.

$$P_n(x) = x^n \left(a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n} \right)$$

$\downarrow$
 ∞

$\xrightarrow{x \rightarrow \infty} o(1)$
 $\xrightarrow{x \rightarrow \infty} o(1)$
 $\underbrace{\hspace{2cm}}_{o(1) \ (x \rightarrow \infty)}$
 $\lim_{x \rightarrow \infty} a_n + o(1) = a_n$

$\downarrow$
无穷大

$$\lim_{x \rightarrow \infty} P_n(x) = \infty.$$

定理. 在自变量的同一变化过程中, 若 $f(x)$ 为无穷大量, 则 $\frac{1}{f(x)}$ 为无穷小量,
 反之, 若 $f(x)$ 为无穷小量, $f(x) \neq 0$, 那么 $\frac{1}{f(x)}$ 为无穷大量.

已经定义了无穷大数列. $\lim_{n \rightarrow \infty} x_n = \infty$.

$\Downarrow$
 当 $x \rightarrow \infty$ 时 Heine 定理.

$$\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{n \rightarrow \infty} x_n \rightarrow A.$$

$$\lim_{n \rightarrow \infty} f(x_n) \rightarrow A$$

证: $\lim_{x \rightarrow \infty} \sin x$ 极限不存在

$$\uparrow \uparrow$$
$$x_n = 2n\pi + \frac{\pi}{2}$$

$$y_n = 2n\pi$$

$$\sin(2n\pi + \frac{\pi}{2}) = 1 \quad n \rightarrow \infty$$

$$\sin(2n\pi) = 0 \quad n \rightarrow \infty$$

$\lim_{x \rightarrow \infty} \cos x$ 极限不存在.

七种极限过程:

$$\lim_{x \rightarrow x_0} f(x)$$

$$\lim_{x \rightarrow x_0+} f(x)$$

$$\lim_{x \rightarrow x_0-} f(x)$$

$$\lim_{x \rightarrow \infty} f(x)$$

$$\lim_{x \rightarrow +\infty} f(x)$$

$$\lim_{x \rightarrow -\infty} f(x)$$

$$\lim_{n \rightarrow \infty} C_n$$

因变量的趋势:

① $\lim_{x \rightarrow \odot} f(x) = A$, $A=0$ 时为无穷小.

② $\lim_{x \rightarrow \odot} f(x) = \infty, +\infty, -\infty$, 无穷大

(函数极限和无穷小的关系):

$$\lim_{x \rightarrow \odot} f(x) = A \iff f(x) = A + o(1) \quad (x \rightarrow \odot)$$

§ 极限运算法则. ≠ 0.

定理: 设 $\lim_{x \rightarrow \infty} f(x) = A$, $\lim_{x \rightarrow \infty} g(x) = B$, 则

$$\textcircled{1} \lim_{x \rightarrow \infty} [f(x) \pm g(x)] = A \pm B = \lim_{x \rightarrow \infty} f(x) \pm \lim_{x \rightarrow \infty} g(x) \quad \left. \vphantom{\lim_{x \rightarrow \infty} [f(x) \pm g(x)]} \right\} \Rightarrow \text{有限}$$

$$\textcircled{2} \lim_{x \rightarrow \infty} [f(x)g(x)] = AB = \lim_{x \rightarrow \infty} f(x) \cdot \lim_{x \rightarrow \infty} g(x)$$

$$\textcircled{3} \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{A}{B} = \frac{\lim_{x \rightarrow \infty} f(x)}{\lim_{x \rightarrow \infty} g(x)} \quad (B \neq 0) \quad / \text{' l'H\^opital'}$$

证: $f(x) = A + \alpha(x) \quad (x \rightarrow \infty) \quad g(x) = B + \beta(x) \quad (x \rightarrow \infty)$

$$[f(x) + g(x)] = [A + \alpha(x)] + [B + \beta(x)] = \underline{(A+B)} + \underbrace{(\alpha(x) + \beta(x))}_{o(1) \quad (x \rightarrow \infty)}$$

$$f(x)g(x) = (A + \alpha(x))(B + \beta(x)) = AB + \underbrace{A\beta + B\alpha + \alpha\beta}_{o(1) + o(1) + o(1) \quad (x \rightarrow \infty)}$$

是无穷小.

$$\frac{f(x)}{g(x)} - \frac{A}{B} = \frac{A + \alpha}{B + \beta} - \frac{A}{B}$$

$$= \frac{B\alpha - A\beta}{B(B + \beta)} = \frac{o(1)}{B^2 + B\beta} \quad (x \rightarrow \infty)$$

$B(B + \beta) = B^2 + B\beta \neq 0$

推论: ① 若 $\lim_{x \rightarrow \infty} f(x)$ 存在, $\lim_{x \rightarrow \infty} g(x)$ 不存在, 则 $\lim_{x \rightarrow \infty} [f(x) \pm g(x)]$ 不存在.

② 若 $\lim_{x \rightarrow \infty} f(x) = A \neq 0$, $\lim_{x \rightarrow \infty} g(x)$ 不存在, 则 $\lim_{x \rightarrow \infty} [f(x)g(x)]$ 不存在.

证: ①. 反证: 假设 $\lim_{x \rightarrow \infty} \underbrace{[f(x) + g(x)]}_{h(x)} = A'$ $g(x)$
||

$$\lim_{x \rightarrow \infty} \underbrace{[f(x) + g(x)]} - \lim_{x \rightarrow \infty} \underbrace{f(x)} = \lim_{x \rightarrow \infty} [f(x) + g(x) - f(x)]$$

$$\textcircled{1}. \lim_{x \rightarrow \infty} \underbrace{[a + g(x)]} = a + \lim_{x \rightarrow \infty} g(x)$$

$$\textcircled{2}. \lim_{x \rightarrow \infty} c g(x) = \underline{\underline{c \lim_{x \rightarrow \infty} g(x)}}$$

推论: ① 若 $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = A$, 且 $\lim_{x \rightarrow \infty} g(x) = 0$, 则 $\lim_{x \rightarrow \infty} f(x) = 0$.

② 若 $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = A \neq 0$, $\lim_{x \rightarrow \infty} f(x) = 0$, 同时有

$f(x)$ 在极限过程的某一邻域上不取 0 值, 则 $\lim_{x \rightarrow \infty} g(x) = 0$.

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} \quad x = \frac{1}{2k\pi}$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \sin x$$

例: 设 $P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ ($a_n \neq 0$) 为 n 次实系数多项式.

$$Q_m(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0 \quad (b_m \neq 0)$$

$$\lim_{x \rightarrow \infty} P_n(x) = \infty.$$

$$\lim_{x \rightarrow c} Q_0 = a_0 \quad f(x) = x \quad \text{单位函数}$$

$$\lim_{x \rightarrow c} P_n(x) =$$

$$\lim_{x \rightarrow c} x = c.$$

$$\lim_{x \rightarrow c} x^n = \lim_{x \rightarrow c} \underbrace{x \cdot \dots \cdot x}_n = \left(\lim_{x \rightarrow c} x \right)^n = c^n.$$

$$\lim_{x \rightarrow c} a_n x^n = a_n c^n$$

$$\lim_{x \rightarrow c} P_n(c) = a_n c^n + a_{n-1} c^{n-1} + \dots + a_1 c + a_0 = P_n(c)$$

$$\lim_{x \rightarrow c} Q_m(c) = Q_m(c) \neq 0$$

$$\text{有理函数. } \lim_{x \rightarrow c} \frac{P_n(x)}{Q_m(x)} = \frac{\lim_{x \rightarrow c} P_n(x)}{\lim_{x \rightarrow c} Q_m(x)} = \frac{P_n(c)}{Q_m(c)}$$

例: 求极限:

$$\textcircled{1} \lim_{x \rightarrow 1} \frac{x^2 + 1}{x^2 + x - 2} = \infty$$

$$\textcircled{2} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + x - 2} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} x^2 + 1 = 2$$

$$x^2 + x - 2 = 0$$

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 + 1} = \frac{0}{2} = 0$$

$$\lim_{x \rightarrow 1} \underbrace{(x^2 + 1)}_{\neq 0} \cdot \left(\frac{1}{x^2 + x - 2} \right) = \infty$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x+2)} = \lim_{x \rightarrow 1} \frac{x+1}{x+2} = \frac{2}{3}$$

例: 求极限:

$$\lim_{x \rightarrow \infty} \frac{x^3 + 1}{x^2 + x - 2}$$

$$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^3 + x - 2}$$

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 1}{x^2 + x - 2}$$

$$\lim_{x \rightarrow \infty} \frac{x(x+1) - 2}{1 + \frac{1}{x^3}} = \infty$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \frac{1}{x^3}}{1 + \frac{1}{x^2} - \frac{1}{x^3}} = 0.$$

$$\lim_{x \rightarrow \infty} \frac{2 - \frac{1}{x^2}}{1 + \frac{1}{x} - \frac{2}{x^2}} = 2$$

一般地, 对于有理函数 (两个多项式的商) 有:

$$\lim_{x \rightarrow \infty} \frac{P_n(x)}{Q_m(x)} = \lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0} = \begin{cases} \infty & n > m \\ \frac{a_n}{b_m} & n = m \\ 0 & n < m \end{cases}$$

$\begin{cases} a_n \neq 0 \\ b_m \neq 0 \end{cases}$

例：求极限：

$$\textcircled{1} \lim_{x \rightarrow \infty} \frac{3x + \sin 2x}{7x + \arctan x}$$

$$\lim_{x \rightarrow \infty} \frac{3 + \frac{\sin 2x}{x} = o(1)}{7 + \frac{\arctan x}{x} = o(1)} = \frac{3}{7}$$

例：求极限：

$$\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{2}{x^2-1} \right)$$

$\lim_{x \rightarrow \infty} \cos x$ 不存在
 $\lim_{n \rightarrow \infty} \cos n$ 不存在
 无穷大 $\uparrow$ 无穷大

$$\textcircled{2} \lim_{n \rightarrow \infty} \frac{5^n + 3^n \cos n}{5^{n+1} + (-2)^n + 1}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{5} + \frac{1}{5} \cdot \underbrace{\left(\frac{3}{5}\right)^n \cos n}_{=o(1)}}{1 + \underbrace{\frac{1}{5} \cdot \left(-\frac{2}{5}\right)^n}_{o(1)} + \underbrace{\frac{1}{5^{n+1}}}_{(1)}} = \frac{1}{5}$$

例: 求常数 a, b , 使得 $\lim_{x \rightarrow 2} \frac{x^2 + ax + b}{x^2 - x - 2} = 2$.

解: $\lim_{x \rightarrow 2} x^2 + ax + b = 4 + 2a + b = 0 \Rightarrow \underline{\underline{a = -2 - \frac{b}{2}}}$

$$\lim_{x \rightarrow 2} x^2 - x - 2 = 4 - 2 - 2 = 0.$$

$$x^2 - (-2 - \frac{b}{2})x + b = (x - 2)(x - \frac{b}{2})$$

$$\lim_{x \rightarrow 2} \frac{(x - 2)(x - \frac{b}{2})}{(x - 2)(x + 1)} = \lim_{x \rightarrow 2} \frac{x - \frac{b}{2}}{x + 1} = \frac{2 - \frac{b}{2}}{3} = 2$$

$$(x - 2)(x + 1)$$

$$b = -8$$

$$a = 2$$

无穷大

不为0

例：求常数 a, b ，使得 $\lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x+1} - (ax+b) \right) = 0$

解： $\lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x+1} - ax \right) = b$

$$\lim_{x \rightarrow \infty} x \left(\frac{x^2+1}{x(x+1)} - a \right) = b.$$

$$\Leftrightarrow \lim_{x \rightarrow \infty} \frac{\frac{x^2+1}{x(x+1)} - a}{\frac{1}{x}} = b$$

$$\lim_{x \rightarrow \infty} \left[\frac{x^2+1}{x(x+1)} - a \right] = 0.$$

$$1 = \lim_{x \rightarrow \infty} \frac{x^2+1}{x(x+1)} = a$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x+1} - x \right) = b$$

$$= \lim_{x \rightarrow \infty} \frac{x^2+1-x^2-x}{x+1} = \lim_{x \rightarrow \infty} \frac{1-x}{x+1} = -1 = b$$

§ 复合函数的极限.

$y = f(u)$
外部函数 内部函数.

定理: 对于复合函数 $f(g(x))$, 设 $\lim_{x \rightarrow x_0} g(x) = u_0$, $\lim_{u \rightarrow u_0} f(u) = A$, 且 $f(g(x))$ 在 x_0 的某去心邻域内有定义, 如果有 $f(u_0) = A$, 则

$$\lim_{x \rightarrow x_0} f(g(x)) = A = f(u_0)$$

证明: $\forall \varepsilon > 0$, $\exists \eta > 0$, s.t. $0 < |x - x_0| < \eta$ $|f(g(x)) - A| < \varepsilon$.

外部函数: $\forall \varepsilon > 0$, $\exists \delta > 0$, s.t. $0 < |u - u_0| < \delta$ 时 $|f(u) - A| < \varepsilon$

条件: $f(u_0) = A$

s.t. $|u - u_0| < \delta$, 有 $|f(u) - A| < \varepsilon$

取 $\delta > 0$, $\exists \eta > 0$, s.t. $0 < |x - x_0| < \eta$ 有 $|g(x) - u_0| < \delta$.

变量代换 $u = g(x)$,

$$\text{有 } |f(u) - A| = |f(g(x)) - A| < \varepsilon$$

例: $f(u) = \begin{cases} 0 & u \neq 0 \\ 1 & u = 0 \end{cases} \quad g(x) = 0$

$f(0) = 1 \neq 0$

$\lim_{x \rightarrow 0} g(x) = 0, \quad \lim_{u \rightarrow 0} f(u) = 0$

$f(g(x)) \equiv 1 \quad \lim_{x \rightarrow 0} f(g(x)) = 1 \neq 0$

定理: 设 $\lim_{x \rightarrow x_0} g(x) = u_0$, $\lim_{u \rightarrow u_0} f(u) = A$, 且 $f(g(x))$ 在 x_0 的某去心邻域内有定义,

如果在 x_0 的某去心邻域内有 $g(x) \neq u_0$, 则 $\lim_{x \rightarrow x_0} f(g(x)) = A$.

$\lim_{x \rightarrow 0^+} \arctan \frac{1}{x} = \frac{\pi}{2} \quad \lim_{x \rightarrow 0^-} \arctan \frac{1}{x} = -\frac{\pi}{2}$

$\lim_{x \rightarrow 0} \arctan \frac{1}{x}$ 不存在

外部已取.

例: 证明 $\lim_{x \rightarrow 0} a^x = 1$ ($a > 0$)

解: 先证 $\lim_{x \rightarrow 0^+} a^x = 1$.

① $a=1$ 显然.

② $a > 1$, $\forall \varepsilon > 0$, $0 < a^x - 1 < \varepsilon$ $\ln a > 0$.

$$\Leftrightarrow 1 < a^x < 1 + \varepsilon$$

$$\Leftrightarrow 0 < x < \frac{\ln(1+\varepsilon)}{\ln a}$$

$$\text{取 } \delta = \frac{\ln(1+\varepsilon)}{\ln a}$$

③ $0 < a < 1$. $b = \frac{1}{a} > 1$.

$$\frac{1}{\lim_{x \rightarrow 0^+} (\frac{1}{a})^x} = \lim_{x \rightarrow 0^+} \frac{1}{b^x} = \frac{1}{1} = 1$$

$$= \lim_{x \rightarrow 0^+} a^x$$

μ μ $\ln a$ $\lim_{x \rightarrow a} x^\mu = e^{\mu \ln x}$

$$= a^\mu$$

$$\lim_{x \rightarrow 0^-} a^x = 1$$

$$x = -t \quad (t = -x)$$

$$x \rightarrow 0^- \quad t \rightarrow 0^+$$

$$\lim_{x \rightarrow 0^-} a^x = \lim_{t \rightarrow 0^+} a^{-t} = \lim_{t \rightarrow 0^+} \frac{1}{a^t} = 1$$

$$\lim_{x \rightarrow b} a^x = a^b \quad \text{对 } \mathbb{R} \text{ 成立}$$

$$a^x = a^b \cdot \underline{a^{x-b}}$$

$$\lim_{x \rightarrow b} a^{x-b} \quad u = x-b \quad x \rightarrow b \quad u \rightarrow 0$$

$$\lim_{x \rightarrow b} a^{x-b} = \lim_{u \rightarrow 0} a^u = 1$$

$$\lim_{x \rightarrow a} \ln x = \ln a \quad (a > 0)$$

对 $a > 0$ 时

$$\lim_{x \rightarrow a} \ln x = \ln a$$

$x^\mu \quad \mu \in \mathbb{Z} \quad \text{对 } \mathbb{R} \text{ 成立}$

$\mu \in \mathbb{R} \setminus \mathbb{Z} \quad [0, +\infty)$

推论: ① 若 $\lim_{x \rightarrow \infty} f(x) = A > 0$, $\lim_{x \rightarrow \infty} g(x) = B$, 则幂指函数 $[f(x)]^{g(x)}$ 极限存在, 且

$$\lim_{x \rightarrow \infty} [f(x)]^{g(x)} = A^B, \quad \lim_{x \rightarrow \infty} f(x)^{\frac{1}{n}} = A^{\frac{1}{n}}$$

② 若 $\lim_{x \rightarrow \infty} f(x) = A > 0$, 则 $\lim_{x \rightarrow \infty} [f(x)]^{\frac{r}{s}} = A^{\frac{r}{s}}$, $r, s \in \mathbb{Z}, (r, s) = 1$

③ 若 $\lim_{x \rightarrow \infty} g(x) = B$, 则 $\lim_{x \rightarrow \infty} e^{g(x)} = e^B$.

证: $f(x)^{g(x)} = e^{g(x) \ln f(x)}$

$$\lim_{x \rightarrow \infty} \ln f(x) = \ln A$$

$$\lim_{x \rightarrow \infty} \underline{\underline{g(x) \ln f(x)}} = B \ln A$$

$$\lim_{x \rightarrow \infty} e^{B \ln A} = A^B$$

$$\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}$$

例: 求极限:

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{(2 + \sin x)^{\frac{3}{2+x \sin \frac{1}{x}}}}{2} = 2^3$$

$$\lim_{x \rightarrow 0} \sin x = 0$$

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$$\textcircled{2} \lim_{n \rightarrow \infty} \frac{n \arctan n}{\sqrt{n^2+1}}$$

keine $\lim_{n \rightarrow \infty} \arctan n = \frac{\pi}{2}$

无意义

$$\lim_{n \rightarrow \infty} \frac{\arctan n}{\frac{\sqrt{n^2+1}}{n}} = \sqrt{1 + \frac{1}{n^2}} = 1 = \frac{\pi}{2}$$

例: 证明 $\lim_{x \rightarrow 1^-} \left(\sqrt{\frac{1}{1-x} + 1} - \sqrt{\frac{1}{1-x} - 1} \right) = 0$

$\frac{1}{1-x} \rightarrow 1^-$ $\downarrow$ $\downarrow$
 ∞ ∞

做变量代换 $y = \frac{1}{1-x}$ $x \rightarrow 1^-$ $y \rightarrow +\infty$

若 $\lim_{y \rightarrow +\infty} (\sqrt{y+1} - \sqrt{y-1}) = 0$, $\Rightarrow$ 原极限 $= 0$.

证: $\lim_{y \rightarrow +\infty} \frac{2}{\sqrt{y+1} + \sqrt{y-1}} = 0$

$\underbrace{\quad \quad \quad}_{+\infty \quad +\infty}$

$$\left| \frac{2}{\sqrt{y+1} + \sqrt{y-1}} \right| < \varepsilon$$

$$\Rightarrow \left| \frac{2}{\sqrt{y-1}} \right| < \varepsilon$$

取 $Y = 1 + \frac{4}{\varepsilon^2} > 0$ 时

§ 三 明治原理

(极限的不等式性质) 若在点 x_0 的某去心邻域内有 $f(x) \geq g(x)$,

$$\lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{x \rightarrow x_0} g(x) = B, \quad \text{则 } A \geq B.$$

定理 (三明治原理) 如果

① 在点 x_0 的某去心邻域内有 $g(x) \leq f(x) \leq h(x)$

② $\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = A$, 则

$$\lim_{x \rightarrow x_0} f(x) = A.$$

证: 有 $g(x) \leq f(x) \leq h(x)$

$\forall x \in \dot{U}_\delta(x_0)$

先设 $g(x) < h(x)$

$$\lambda(x) = \frac{f(x) - g(x)}{h(x) - g(x)} \in [0, 1].$$

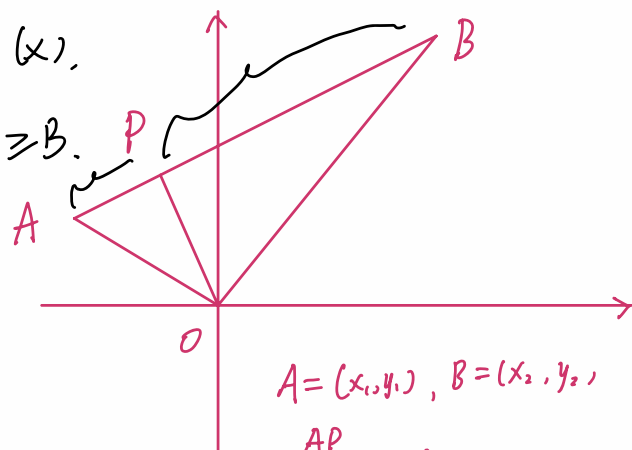
若 $\lambda(x) = 0$, $g(x) = h(x)$

$\lambda(x)$ 也是 $\dot{U}_\delta(x_0)$ 上的值域 $[0, 1]$

$$f(x) = \lambda(x)g(x) + (1 - \lambda(x))h(x)$$

$$\begin{cases} g(x) = A + \alpha(x) \\ h(x) = A + \beta(x) \end{cases} \quad (x \rightarrow x_0)$$

定比分点公式:



$$A = (x_1, y_1), B = (x_2, y_2)$$

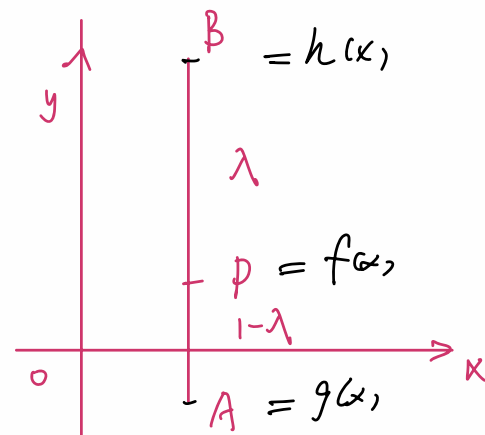
$$\frac{AP}{PB} = \alpha$$

设 $P = (x, y)$

$$\text{则 } x = \frac{x_1 + \alpha x_2}{1 + \alpha}, \quad y = \frac{y_1 + \alpha y_2}{1 + \alpha}$$

$$f(x) = A + [\lambda(x)\alpha(x) + (1 - \lambda(x))\beta(x)]$$

$\downarrow$
 $o(1) \quad (x \rightarrow x_0)$



$$\alpha = \frac{1 - \lambda}{\lambda}$$

$$y = \lambda y_1 + (1 - \lambda) y_2$$

例: 求极限 $\lim_{n \rightarrow \infty} \left(\frac{1}{n^2+1} + \frac{1}{n^2+2} + \dots + \frac{1}{n^2+n} \right)$

记为 $A(n)$

最大的 $\frac{1}{n^2+1}$ 最小的 $\frac{1}{n^2+n}$

$$\frac{1}{n^2+n} (1 + \dots + n) \leq A(n) \leq \frac{1}{n^2+1} (1 + \dots + n)$$

$$\frac{n(n+1)}{2(n^2+n)} \leq A(n) \leq \frac{n(n+1)}{2(n^2+1)}$$

$$\lim_{x \rightarrow +\infty} \frac{x(x+1)}{2(x^2+1)} = \frac{1}{2} \quad \Rightarrow \quad \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2+1)} = \frac{1}{2}$$

$$\lim_{x \rightarrow +\infty} \frac{x(x+1)}{2(x^2+x)} = \frac{1}{2} \quad - \quad - \quad - \quad = \frac{1}{2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

例: 证明 $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$, $a > 0$ 为常数

证: ① $a=1$, 显然

② $a > 1$ 情况. $\sqrt[n]{a} > 1$, 设 $\sqrt[n]{a} = 1 + \lambda_n$, $\lambda_n > 0$.

$$a = (1 + \lambda_n)^n = 1 + C_n^1 \lambda_n + \underbrace{C_n^2 \lambda_n^2 + \dots + C_n^n \lambda_n^n}_{\geq C_n^1 \lambda_n}$$

$$C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!} \quad C_n^1 = n \quad \text{由 } \lambda_n > 0$$

$$\geq 1 + n \lambda_n \quad \text{证者利证.}$$

$$0 < \lambda_n < \frac{a-1}{n}$$

$$\lim_{n \rightarrow \infty} 0 = 0 \quad \lim_{n \rightarrow \infty} \lambda_n = 0. \quad \lim_{n \rightarrow \infty} \frac{a-1}{n} = 0.$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = \lim_{n \rightarrow \infty} (1 + \lambda_n) = 1 + \lim_{n \rightarrow \infty} \lambda_n = 1 + 0 = 1.$$

③ $0 < a < 1$ $b = \frac{1}{a} > 1$. $\lim_{n \rightarrow \infty} \sqrt[n]{b} = 1$

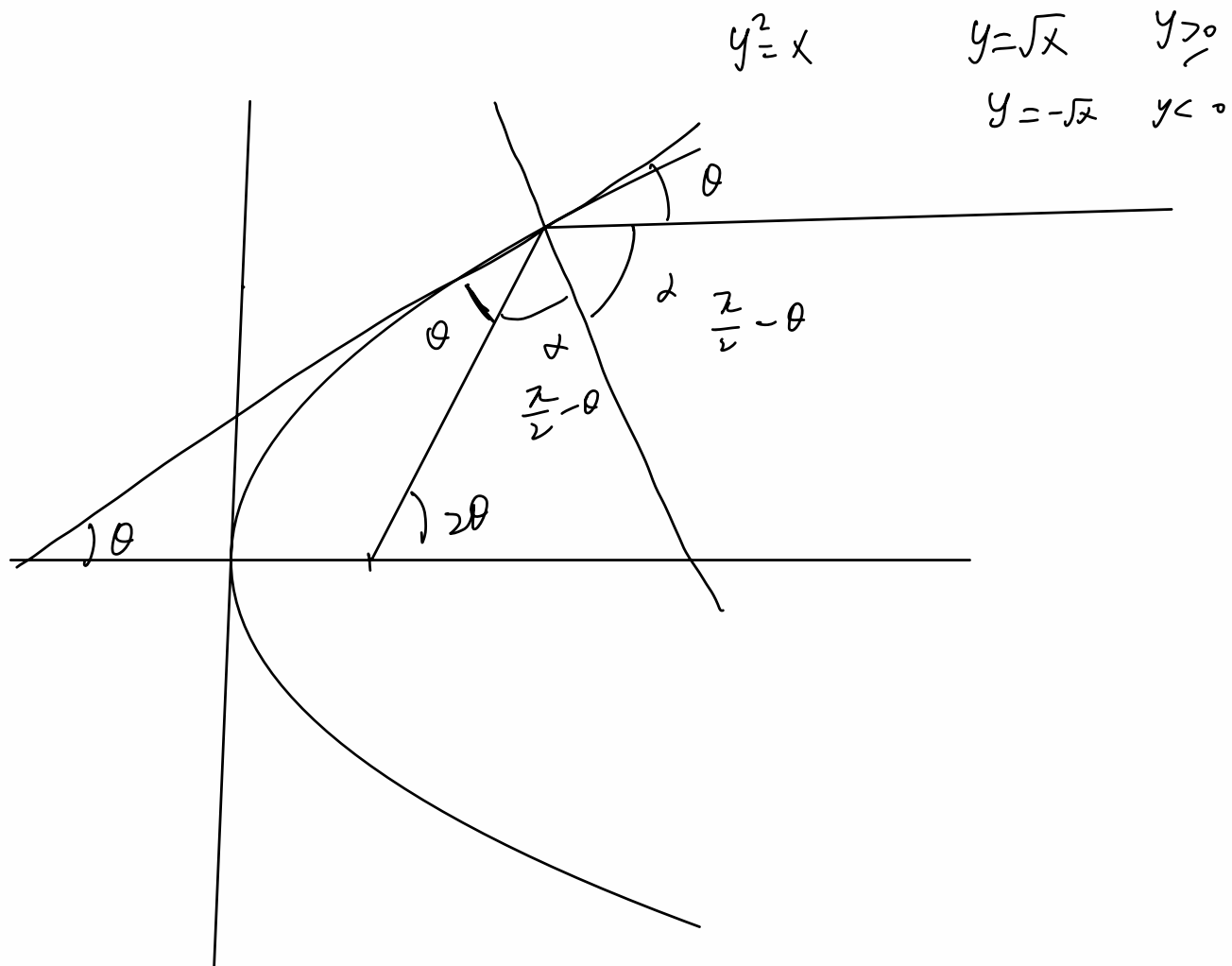
$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{b}} = \frac{1}{1} = 1$$

① 例: 求极限 $\lim_{n \rightarrow \infty} \sqrt[n]{2^n + 3^n + 7^n}$

$$\sqrt[n]{2^n} \quad \sqrt[n]{\frac{2^n + 2^n + 2^n}{2}} \quad A(n)$$

$$\begin{array}{c} \sqrt[n]{7^n} \\ \parallel \\ 7 \end{array} \leq \underline{\underline{A(n)}} \leq \sqrt[n]{7^n + 7^n + 7^n}$$
$$= 7 \cdot \begin{array}{c} \sqrt[n]{3} \\ \parallel \\ 1 \end{array}$$

② $\lim_{x \rightarrow 0^+} x \left[\frac{1}{x} \right]$ (取整函数)



导数与微分

极限 . 函数的变化趋势.

自变量变化的快慢. 变化率. 一个变化过程中改变的平均值.

导数 微分

直线斜率: $y=f(x)$. $(x_0, f(x_0))$ $k = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

瞬时速度

设一个质点做非匀变速运动

$$s = s(t).$$

$$\bar{v}(\Delta t) = \frac{\Delta s}{\Delta t} = \frac{s(t_0 + \Delta t) - s(t_0)}{\Delta t}$$

$$s = \frac{1}{2}gt^2 \quad \begin{array}{ll} t=0 & \text{到 } t=1 \\ t=1 & \text{到 } t=2 \end{array}$$

瞬时 $v(t)$ (连续)

$$\boxed{\Delta t \rightarrow 0}$$

$$v(t_0 + \Delta t) - v(t_0) \rightarrow 0.$$

$$v(t_0) = \lim_{\Delta t \rightarrow 0} \bar{v}(\Delta t) = \lim_{\Delta t \rightarrow 0} \frac{s(t_0 + \Delta t) - s(t_0)}{\Delta t}$$

$\lim_{\Delta x \rightarrow 0} \frac{\text{函数的改变量 (增量)}}{\text{自变量的改变量}}$ 的极限

定义: 设函数 $f(x)$ 在 x_0 的邻域上有定义, 任给 x_0 处一个变化量 (增量) $\Delta x \neq 0$, ($\Delta x > 0, \Delta x < 0$).

相应的函数增量 $\Delta y = f(x_0 + \Delta x) - f(x_0)$. 若极限

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \text{ 极限存在.}$$

把 x_0 换作 $x \in (a, b)$

则称函数 $f(x)$ 在 x_0 处可导的, 把这极限记为 $f(x)$ 在 x_0 的导数.

$$\frac{df(x)}{dx} \Big|_{x=x_0}, \quad \frac{dy}{dx} \Big|_{x=x_0}, \quad \boxed{f'(x_0)}, \quad y' \Big|_{x=x_0}.$$

①.

$$x = x_0 + \Delta x \quad f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad \text{导数}$$

函数曲线 $y=f(x)$. 在 $(x_0, f(x_0))$ $k = f'(x_0)$

切线

法线

$$k = f'(x_0) \neq 0$$

$$k = 0$$

$$k = \infty$$

$$y - f(x_0) = f'(x_0)(x - x_0)$$

$$y = f(x_0)$$

$$y - f(x_0) = -\frac{1}{k}(x - x_0)$$

$$x = x_0$$

若 ① 极限不存在, 我们称 $f(x)$ 在 x_0 处不可导的.

-- 但是无穷大, 记为 $f'(x_0) = \infty$.

$$y \quad f(y) - f(x_0)$$

例: 设 $f'(0) = A$, 求 $\lim_{x \rightarrow 0} \frac{f(e^{2x} - 1) - f(0)}{\sin x} = 2A$

解: $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = A$

$$\lim_{x \rightarrow 0} \frac{f(e^{2x} - 1) - f(0)}{e^{2x} - 1} \cdot \frac{e^{2x} - 1}{\sin x} \sim \frac{\frac{0}{0}}{\frac{2x}{x}} = 2$$

$$y = e^{2x} - 1 \quad x \rightarrow 0 \quad y \rightarrow 0$$

$$\lim_{y \rightarrow 0} \frac{f(y) - f(0)}{y} = A$$

$$f(x_0) \quad x_0 + (-\Delta x)$$

例: 设 $f'(x_0)$ 存在, 求 $\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{\Delta x}$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0) + f(x_0) - f(x_0 - \Delta x)}{\Delta x}$$

$$f(x_0 + \Delta x) - f(x_0) \quad \lim_{\Delta x \rightarrow 0} \frac{f(x_0 - \Delta x) - f(x_0)}{-\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{f(x_0) - f(x_0 - \Delta x)}{-\Delta x}$$

$$= f'(x_0) + f'(x_0) = 2f'(x_0)$$

$f(x)$ 在 x_0 的左 δ -邻域 $(x_0 - \delta, x_0]$ 上有定义.

(单侧导数)

定义: 若极限

$$\lim_{\Delta x \rightarrow 0^-} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{极限存在}$$

$$\lim_{\Delta x \rightarrow 0^+} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} \quad \text{极限存在}$$

$f(x)$ 在 x_0 的右 δ -邻域 $[x_0, x_0 + \delta)$ 上有定义.

称极限值 $f'(x_0)$ 在 x_0 处的左 (右) 导数.

$$f'_-(x_0), \quad f'_+(x_0).$$

定理: 导数 $f'(x_0)$ 存在 $\Leftrightarrow f'_-(x_0) = f'_+(x_0)$.

例: 求函数 $f(x) = \begin{cases} \sin x & x < 0 \\ \ln(1+x) & x \geq 0 \end{cases}$ 在 $x=0$ 处的导数.

解: $f(0)=0$.

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0^-} \frac{\sin x}{x} = 1.$$

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{\ln(1+x)}{x} = 1 \Rightarrow f'(0) = 1.$$

函数 $f(x)$ 在开区间 (a, b) 中的每一个点都是可导的, $\forall x \in (a, b)$.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{(h)} = f'(x), \text{ 存在.}$$

则称 $f(x)$ 在 (a, b) 上可导, $f'(x)$ 也称作 $f(x)$ 在 (a, b) 上的导函数.

闭区间 $[a, b]$, $f(x)$ ① 在 (a, b) 上可导

② 在 a 处左可导

③ 在 b 处右可导

$\Rightarrow f(x)$ 在 $[a, b]$ 上可导

$$\underline{f'(x)}, \quad \frac{df(x)}{dx}, \quad y'(x), \quad \frac{dy(x)}{dx}$$

$$\frac{df}{dx} \text{ 或 } y' \text{ 或 } \frac{dy}{dx}$$

$$\text{在点 } x_0 \in (a, b), \quad f'(x_0) = f'(x) \Big|_{x=x_0}.$$

也称作导函数是导数

可导 $\Rightarrow$ 连续. 连续性是可导性的必要条件.

定理: 若函数 $f(x)$ 在 x_0 处可导, 那么函数 $f(x)$ 在 x_0 处连续的.

证明: 取 $f'(x_0)$ 存在 $\in \mathbb{R}$ (常数)

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \quad (\text{极限和无穷小关系}).$$

$$\Leftrightarrow \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) + o(1) \quad (x \rightarrow x_0)$$

$$\Leftrightarrow f(x) - f(x_0) = \underbrace{f'(x_0)}_{o(1)} \underbrace{(x - x_0)}_{o(1)} + \underbrace{o(1)(x - x_0)}_{o(1)} = o(1) \quad (x \rightarrow x_0)$$

$$\Leftrightarrow \lim_{x \rightarrow x_0} [f(x) - \boxed{f(x_0)}] = 0.$$

$$\Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0) \quad (\text{连续性的定义})$$

逆否命题: $f(x)$ 在 x_0 处不连续 $\Rightarrow f(x)$ 在 x_0 处不可导.

$f(x)$ 连续性 $\nleftrightarrow$ 可导性.

连续函数不可导的几种情况.

可导性 $\Leftrightarrow \frac{f(x) - f(x_0)}{x - x_0}$ 极限存在性.

① 左、右极限虽然存在但不相等 $f'_-(x_0) \neq f'_+(x_0)$

② 在右极限中至少一个不存在 $f'_-(x_0)$ 或 $f'_+(x_0)$ 不存在.

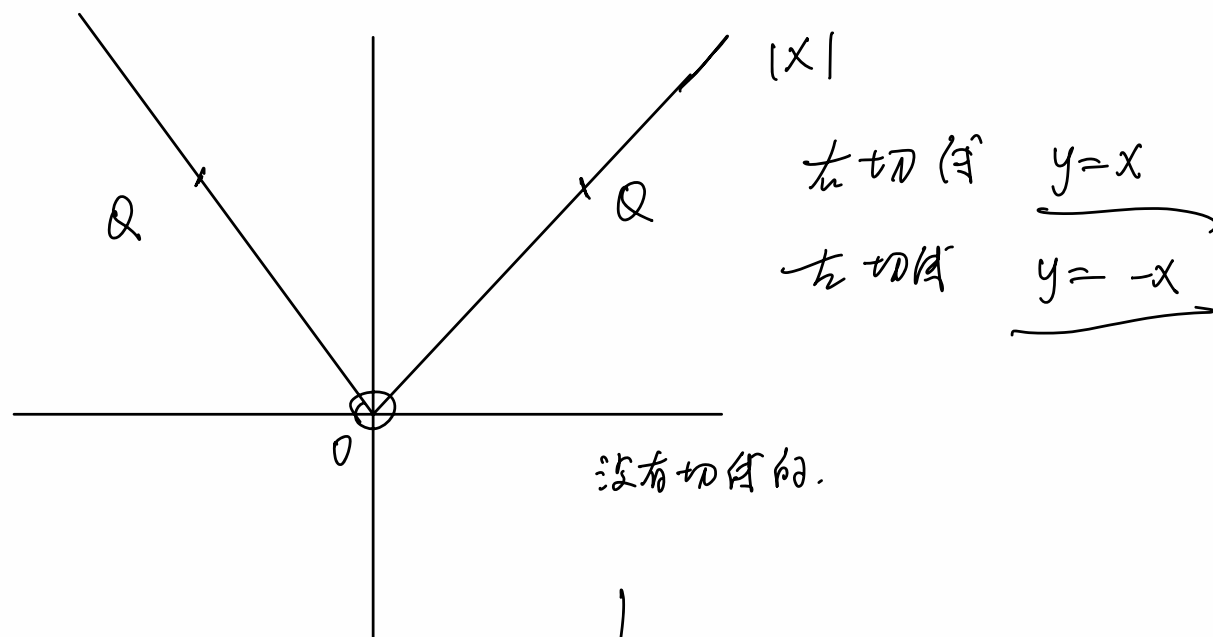
③ $f'_-(x_0)$ 或 $f'_+(x_0)$ 至少一个 $\neq \infty$.

例: $f(x) = |x|$ $(-\infty, +\infty)$ $= \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$ 在 $x=0$ 处不可导.
 $|x| = \sqrt{x^2}$ 连续性 $\checkmark$.

$$f(0) = 0.$$

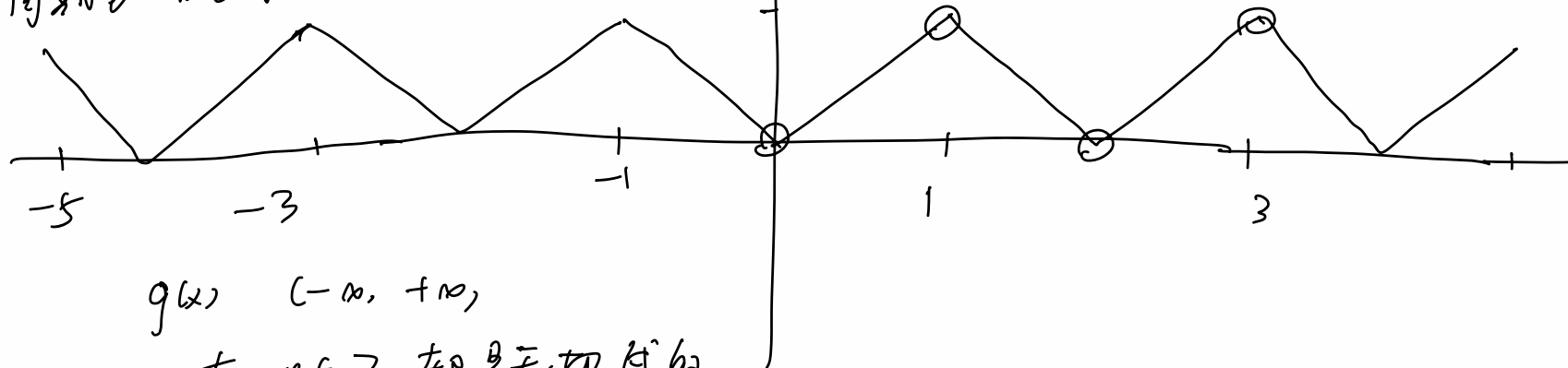
$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x)}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$$

$$f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x)}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$



$|x|$ 在 $[-1, 1]$.

做周期为2的延拓



$g(x) \quad (-\infty, +\infty)$

在 $n \in \mathbb{Z}$ 都是无切线的

(Weierstrass 例子)

例: $f(x) = 2\sqrt{x}$ 在 $(-\infty, +\infty)$ 上连续, 但是在 $x=0$ 处不可导
偶函数

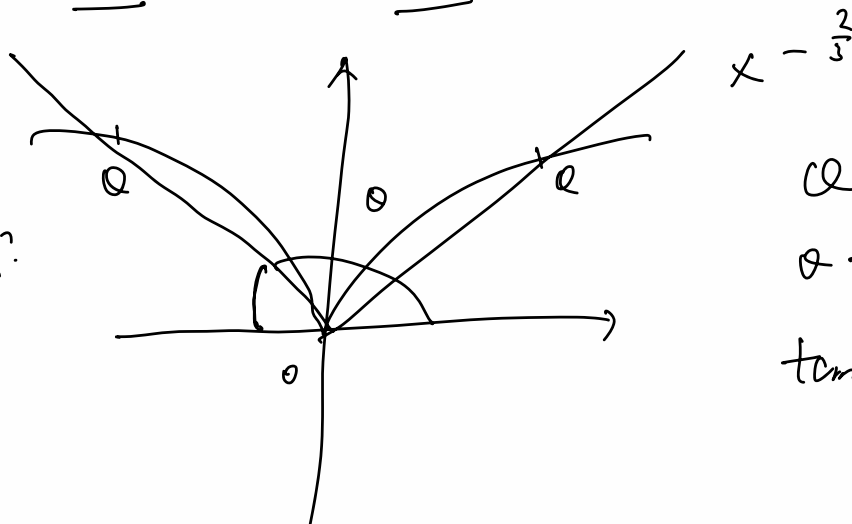
$$f(0) = 0.$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{x^{\frac{1}{2}}}{x} = \lim_{x \rightarrow 0} \frac{1}{x^{\frac{1}{2}}} = +\infty.$$

(几何)

$x=0$ 是在 $x=0$ 处的切线?

不可导 但是切线存在



$$\theta \rightarrow 0, -1$$

$$\theta \rightarrow \frac{\pi}{2}$$

$$\tan \theta \rightarrow +\infty$$

例: $f(x) = \begin{cases} x \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$ $x=0$ 处不可导.

解: $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 = f(0)$ 连续的.

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0^+} \frac{x \sin \frac{1}{x}}{x} = \lim_{x \rightarrow 0^+} \sin \frac{1}{x} \text{ 不存在}$$

$\lim_{x \rightarrow 0^-} \frac{f(x)}{x}$ 也不存在

定理: 函数 $f(x)$ 在 x_0 处左(右)导数存在, 则 $f(x)$ 在 x_0 处左(右)连续.

$f(x)$ 在 x_0 处左可导且右可导, 且 $f'_-(x_0) = f'_+(x_0)$ 则 $f(x)$ 在 x_0 处可导.

例: 函数 $f(x) = \begin{cases} e^x & x \leq 2 \\ ax+b & x > 2 \end{cases}$ 设 $f'(2)$ 存在, 求 a 和 b 的值.

解: $f'(2)$ 存在 $\Rightarrow f(x)$ 在 2 处连续.

$$\lim_{x \rightarrow 2} f(x) = f(2).$$

$$\lim_{x \rightarrow 2^-} e^x = (e^2) = \lim_{x \rightarrow 2^+} ax+b = \boxed{2a+b} = f(2)$$

左导数:

$$f'_-(2) = \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2}$$

$$= \lim_{x \rightarrow 2^-} \frac{e^x - e^2}{x - 2} = e^2 \left[\lim_{x \rightarrow 2^-} \frac{e^{x-2} - 1}{x - 2} \right] = e^2$$

右导数 = 左导数.

$$f'_+(2) = \lim_{x \rightarrow 2^+} \frac{ax + b - f(2)}{x - 2} = \lim_{x \rightarrow 2^+} \frac{a(x-2)}{x-2} = a.$$

$$e^2 = a$$

$$e^2 = 2a + b$$

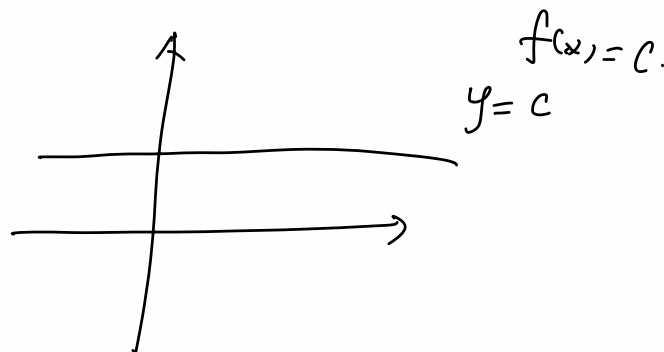
$$e^2 = 2e^2 + b \Rightarrow b = -e^2$$

从导数的定义出发求连续函数的导数 (导函数)

① 常值函数 $f(x) = c$ $c \in \mathbb{R}$ $\forall x \in (-\infty, +\infty)$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0.$$

$$\text{即 } (c)' = 0.$$



② 正弦函数 $f(x) = \sin x$ $x \in (-\infty, +\infty)$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos(x + \frac{h}{2}) \sin \frac{h}{2}}{h} \quad h \rightarrow 0 \quad \frac{h}{2} \rightarrow 0 \\ &= \lim_{h \rightarrow 0} \underbrace{2 \cos(x + \frac{h}{2})}_{\cos x} \underbrace{\frac{\sin \frac{h}{2}}{\frac{h}{2}}}_{\sim \frac{1}{2}} = \cos x \end{aligned}$$

$$\underline{\underline{(\sin x)' = \cos x}}$$

$$(\cos x)' = -\sin x$$

$$\left\{ \begin{array}{l} \tan x \\ \arcsin x \end{array} \right.$$

③ 幂函数 $f(x) = x^\alpha$ ($\alpha \neq 0$), $\frac{1}{x} \rightarrow 0$ 时的导数

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^\alpha - \boxed{x^\alpha}}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^\alpha \left[\left(1 + \frac{h}{x}\right)^\alpha - 1 \right]}{h} \quad h \rightarrow 0 \quad \left(\frac{h}{x} \rightarrow 0 \right) \\ &= x^\alpha \underbrace{\lim_{h \rightarrow 0} \frac{\left(1 + \frac{h}{x}\right)^\alpha - 1}{h}}_{\frac{\alpha}{x}} \sim \frac{\alpha \cdot \frac{h}{x}}{h} \end{aligned}$$

$$= 2x^{\alpha-1}$$

$$(x^\alpha)' = 2x^{\alpha-1}$$

$$(x)' = 1$$

$$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$$

$$(\sqrt{x})' = \frac{1}{2\sqrt{x}} \quad (x > 0)$$

④ 指数函数 $f(x) = a^x$ ($a > 0, a \neq 1$)

$\ln a$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{a^{x+h} - \underline{a^x}}{h} = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} \\ &= a^x \ln a. \end{aligned}$$

$$(a^x)' = a^x \ln a.$$

取 $a = e$.

$$(e^x)' = e^x$$

⑤. 对数函数 $f(x) = \log_a x$ ($a > 0, a \neq 1$) ($0, +\infty, \forall x \in (0, +\infty)$).

$$f'(x) = \lim_{h \rightarrow 0} \frac{\log_a (x+h) - \log_a x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\log_a \left(1 + \frac{h}{x}\right)}{\frac{h}{x}}$$

$$= \lim_{h \rightarrow 0} \frac{\ln \left(1 + \frac{h}{x}\right)}{\ln a} \cdot \frac{1}{h}$$

$$= \frac{1}{\ln a} \lim_{h \rightarrow 0} \frac{\ln \left(1 + \frac{h}{x}\right)}{\frac{h}{x}} \sim \frac{h}{x} = \frac{1}{x}$$

$$= \frac{1}{x \ln a}$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

$$a = e.$$

$$(\ln x)' = \frac{1}{x}$$

例: 设 $f(x)$ 在 $(-\infty, +\infty)$ 上有定义, 且 $\forall x, y$, 满足

$$f(x+y) = f(x)g(y) + f(y)g(x).$$

$$其 g(x) = e^{\sin x} - x \cos x,$$

$$\text{又 } \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1.$$

$$\text{求 } f(0), f'(0), f'(x).$$

解: $g(x)$ 在 $(-\infty, +\infty)$ 上连续.

$$g(0) = 1$$

$$\text{令 } f = y \Rightarrow \quad \underline{f(0)} = f(0)g(0) + f(0)g(0) = 2f(0)g(0).$$

$$\Rightarrow \underline{f(0) = 0.}$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1.$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)g(h) + f(h)g(x) - f(x)}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{g(h) - 1}{h} + g(x) \lim_{h \rightarrow 0} \frac{f(h)}{h} = f(x) \lim_{h \rightarrow 0} \frac{g(h) - 1}{h} + g(x) \underline{\underline{f'(0) = 1}} = g(x)$$

$$g'(0) = \lim_{h \rightarrow 0} \frac{e^{\sinh h} - h \cosh h - 1}{h} = \lim_{h \rightarrow 0} \left(\frac{e^{\sinh h} - 1}{h} - \cosh h \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{\sinh h}{h} - \cosh h \right) = 1 - 1 = 0.$$

$$\underline{\underline{f'(x) = g(x).}}$$

例: 设函数 $f(x)$ 在 $(-\infty, +\infty)$ 上可导, 且满足.

$$f(1+x) - 2f(1-x) = 3x + o(x) \quad (x \rightarrow 0 \text{ 时的高阶无穷小})$$

求曲线 $y=f(x)$ 在点 $x=2$ 处的切线.

解: $f(1)$, $f'(1)$.

$(-\infty, +\infty)$

$f'(1)$ 存在 $\Rightarrow f(x)$ 在 1 点连续.

$$\lim_{h \rightarrow 0} f(1+h) = f(1).$$

$$\lim_{x \rightarrow 0} [f(1+x) - 2f(1-x)] = f(1) - 2f(1) = -f(1) = 0.$$

$$\Rightarrow f(1) = 0.$$

$$\lim_{h \rightarrow 0} \frac{f(1+h)}{h}$$

$$\frac{f(1+x) - 2f(1-x)}{x} = 3 + \frac{o(x)}{x} = o(1)$$

$$\lim_{h \rightarrow 0} \left[\frac{f(1+h)}{h} + \frac{2f(1-h)}{-h} \right] = 3$$

$\downarrow$
 h

$$\frac{f'(1) + 2f'(1)}{1} = 3 \Rightarrow f'(1) = 1$$

$$f(1) = 0$$

$$f'(1) = 1.$$

$$y - 0 = 1(x - 1)$$

$$y = x - 1$$

例: 2.1.10. 假设当 $h \rightarrow 0$, $\alpha(h)$ 和 $\beta(h)$ 是同阶的无穷小, 且 $\alpha(h)$ 在 $h \rightarrow 0$ 邻域内连续且单调, 证明 $f(x)$ 在 a 处可导 $\Leftrightarrow \lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\beta(h)} = A$ 存在.

解: 左 $\Rightarrow$ 右. $f'(a)$ 存在.

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ 存在}$$

$$\lim_{h \rightarrow 0} \frac{\alpha(h)}{\beta(h)} = k \neq 0$$

$\beta(h)$ 在 $h \rightarrow 0$ 邻域均 $\neq 0$.

$$\lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\alpha(h)}$$

$$\frac{\alpha(h)}{\beta(h)} = \frac{1}{\frac{\beta(h)}{\alpha(h)}} = \frac{1}{k}$$

$$\lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\alpha(h)}$$

$$s = \alpha(h)$$

$$h \rightarrow 0, \alpha(h) \rightarrow 0$$

$$\lim_{s \rightarrow 0} \frac{f(a+s) - f(a)}{s} = f'(a)$$

右 $\Rightarrow$ 左.

$$\lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\beta(h)} = A$$

$$\lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\alpha(h)} = A$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{\alpha(h)} = \frac{1}{k}$$

$f' = \lim_{s \rightarrow 0} \frac{f(a+s) - f(a)}{s}$
 F
 $s = \alpha(h)$
 $h = \alpha^{-1}(s)$
 $s \rightarrow 0, h \rightarrow a$
 $S = \alpha(h)$
 $h = \alpha^{-1}(s)$
 $F \circ \alpha$

$\alpha(h)$ 在 $h \rightarrow 0$ 邻域内单调, $h = \alpha^{-1}(t)$ 存在.

$$\lim_{h \rightarrow 0} \alpha(h) = \alpha(0) = 0.$$

$$\alpha^{-1}(0) = 0.$$

$$s = \alpha(h)$$

$$h = \alpha^{-1}(s)$$

$\alpha^{-1}(h)$ 连续. $\lim_{h \rightarrow 0} \alpha^{-1}(h) = \alpha^{-1}(0) = 0.$

$$\alpha(\alpha^{-1}(s))$$

$$\lim_{s \rightarrow 0} \frac{f(a+s) - f(a)}{s} = \lim_{s \rightarrow 0} \frac{f(a + \alpha(\alpha^{-1}(s))) - f(a)}{\alpha(\alpha^{-1}(s))}$$

$$h = \alpha^{-1}(s)$$

$$= \lim_{h \rightarrow 0} \frac{f(a + \alpha(h)) - f(a)}{\alpha(h)} = \frac{1}{k}$$

§ 微分

$$S = x^2$$

$$S' = (x + \Delta x)^2$$

$$\Delta S = S' - S = (x + \Delta x)^2 - x^2$$

$$= \boxed{2x \Delta x} + \underline{\underline{\Delta x^2}}$$

线性主部

x 是常数

$\Delta x \rightarrow 0$

高阶无穷小.

$$\Delta x^2 = o(\Delta x) \quad (\Delta x \rightarrow 0)$$



(a, b)
 $[a, b]$

定义: 设函数 $y = f(x)$ 在区间 $I = (a, b)$ 内有定义, 且 $x_0 \in I = (a, b)$. 当自变量 x 在点 x_0 处有增量 Δx , 且 $x_0 + \Delta x \in I$ 时, 若函数在 x_0 处的增量 Δy 可以表示为

$$\textcircled{1} \text{式} \quad \Delta y = \underline{f(x_0 + \Delta x) - f(x_0)} = \underline{A(x_0) \Delta x} + \underline{\beta(\Delta x)} \quad \text{余量}$$

其中 $A(x_0)$ 是与 Δx 无关的常数, 且当 $\Delta x \rightarrow 0$ 时 $\beta(\Delta x) = o(\Delta x)$ (高阶无穷小),

则称函数 $f(x)$ 在 x_0 处可微; 函数增量的线性主要部分 (简称线性主部)

$A(x_0) \Delta x$ 称为函数 $f(x)$ 在 x_0 处的微分, 记作 $df(x_0)$ 或 $dy|_{x=x_0}$, 即

$$\underline{df(x_0) = dy|_{x=x_0} = A(x_0) \Delta x.}$$

证: Δx 不一定很小.

$$\frac{A(x_0) \Delta x}{\Delta x} \quad \Delta x \rightarrow 0 \quad \text{同阶}$$

$$\Delta y - dy = o(\Delta x) \quad (\Delta x \rightarrow 0)$$

定理: 函数 $y=f(x)$ 在点 x_0 处可微的充分必要条件是函数 $y=f(x)$ 在 x_0 处可导.

$$\text{且 } dy|_{x=x_0} = f'(x_0) \Delta x$$

证明: 先假设 $f(x)$ 在 x_0 处可微.

$$\Delta y = f(x_0 + \Delta x) - f(x_0) = \boxed{A(x_0)} \Delta x + o(\Delta x) \quad (\Delta x \rightarrow 0)$$

$$(\Delta x \neq 0) \quad \frac{\Delta y}{\Delta x} = A(x_0) + \frac{o(\Delta x)}{\Delta x} \quad (\Delta x \rightarrow 0)$$
$$\underline{\underline{= o(1)}}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = A(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

假设函数 $f(x)$ 可导.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$$

$$\Leftrightarrow \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0) + o(1) \quad (\Delta x \rightarrow 0)$$

$$\Leftrightarrow \Delta y = f(x_0 + \Delta x) - f(x_0) = f'(x_0) \Delta x + \underbrace{o(1) \cdot \Delta x}_{\underline{\underline{o(\Delta x)}}} \quad (\Delta x \rightarrow 0)$$

可导与可微.

$$\text{可导: } \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = A = f'(x),$$

$$\text{可微: } \underline{\underline{\Delta y}} = \underline{\underline{f'(x_0) \Delta x}} + o(\Delta x),$$

$$\text{记 } x = x_0 + \Delta x.$$

$$\Delta y = f'(x_0) \Delta x + o(\Delta x)$$

$$\Rightarrow f(x) - f(x_0) = f'(x_0)(x - x_0) + o(x - x_0) \quad (x \rightarrow x_0)$$

$$\Rightarrow \underline{\underline{f(x)}} = \underline{\underline{f(x_0) + f'(x_0)(x - x_0)}} + \underline{\underline{o(x - x_0)}} \quad (x \rightarrow x_0)$$

f 在点 x_0 可导 $\Rightarrow$ 可微.

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

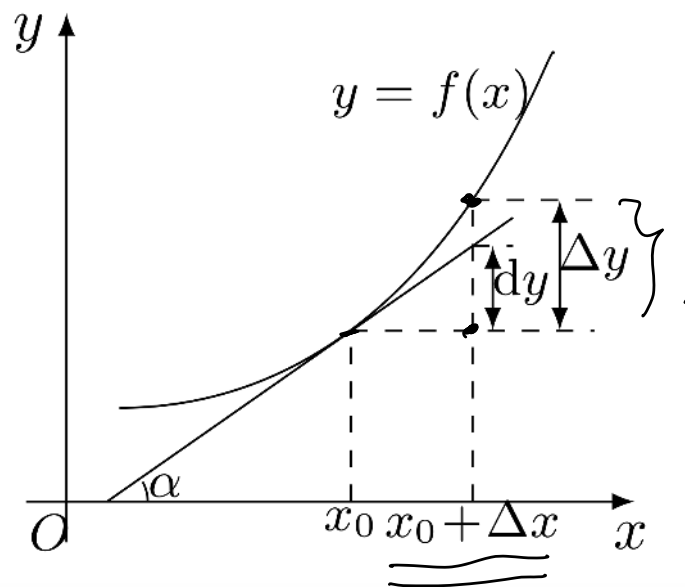
$$\underline{\underline{f(x) = f(x_0) + o(1)}} \quad (x \rightarrow x_0)$$

$$k = f'(x_0)$$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

$$k \cdot \Delta x = dy|_{x=x_0}.$$

$$\Delta y$$



定义: 若 $f(x)$ 在 (a, b) 内每一点都可微, 则 $f(x)$ 是 (a, b) 上的可微函数.
 $[a, b]$.

$$df(x) = f'(x) \Delta x$$

单位映射: $f(x) = x$

$$df(x) = dx = (x)' \Delta x = \Delta x$$

$$dx = \Delta x$$

$$dy = df(x) = f'(x) \Delta x = \boxed{f'(x) dx}$$

① x
 ② dx } 对应的.

$$\frac{dy}{dx} = \frac{df(x)}{dx} = f'(x) \quad (\text{微分})$$

$$y = \sin x \quad (y)' = \cos x \quad dy = \cos x \, dx$$

例: 计算 $\sin 30^\circ 5'$

$$\frac{\pi}{180} \cdot \frac{1}{12}$$

$$30^\circ 5' = \frac{\pi}{6} + \frac{\pi}{216}$$

例: 求 $\sqrt[6]{65}$ 的近似值:

$$\sqrt[6]{65} = \sqrt[6]{64+1} = 2 \sqrt[6]{1+\frac{1}{64}}$$

$$\frac{dy}{dx} = \cos x$$

$$dy = \cos x \, dx$$

$$\Delta y = \sin 30^\circ 5' - \sin \frac{\pi}{6} = \sin 30^\circ 5' - \frac{1}{2}$$

$$dy = \cos x_0 \cdot \Delta x$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{\pi}{2160}$$

$$\Delta y \sim dy$$

$$\sin 30^\circ 5' \sim \sin \frac{\pi}{6} + dy$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{2160}$$

$$f(x) = 2 \cdot \sqrt[6]{x} \quad \text{at } x_0 = 1$$

$$\Delta x = \frac{1}{64}$$

$$f'(x) = \frac{1}{3} x^{-\frac{5}{6}}$$

$$f'(1) = \frac{1}{3}$$

$$f(1) = 2$$

$$\Delta y \sim dy = f'(x) \Delta x = \frac{1}{3} \cdot \frac{1}{64}$$

$$\sqrt[6]{65} - 2 \sim dy = \frac{1}{3} \cdot \frac{1}{64}$$

$$\sqrt[6]{65} \sim \underline{\underline{2 + \frac{1}{192}}}$$

§ 微分法则

定理(四则运算法则) 设函数 $f(x)$ 和 $g(x)$ 在点 x 处可导, 则 $f(x) \pm g(x)$ 和 $f(x)g(x)$ 也在点 x 处可导;
若 $g(x) \neq 0$, 则 $\frac{f(x)}{g(x)}$ 也在点 x 处可导, 且有如下公式:

① (加法法则) $(f(x) \pm g(x))' = f'(x) \pm g'(x)$

$$d(f(x) \pm g(x)) = df(x) \pm dg(x)$$

证明:
$$\begin{aligned} \underline{\underline{(f(x) \pm g(x))'}} &= \lim_{h \rightarrow 0} \frac{[f(x+h) \pm g(x+h)] - [f(x) \pm g(x)]}{h} \quad \text{存在} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)] \pm [g(x+h) - g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \pm \frac{g(x+h) - g(x)}{h} \\ &= f'(x) \pm g'(x) \end{aligned}$$

$$f(x) = x^2 + \sin x$$

$$\begin{aligned} f'(x) &= (x^2)' + (\sin x)' \\ &= 2x + \cos x \end{aligned}$$

$$f(x) = x^3 - \sin x + \ln x + 5$$

$$\begin{aligned} f'(x) &= (x^3)' - (\sin x)' + (\ln x)' + (5)' \\ &= 3x^2 - \cos x + \frac{1}{x} \end{aligned}$$

② (乘法法则) $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$
 $d(f(x)g(x)) = g(x)df(x) + f(x)dg(x)$

证明: $(f(x)g(x))' = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$$

$g'(x) \Rightarrow g(x) \in \mathbb{R}$
 $\lim_{h \rightarrow 0} g(x+h) = g(x)$

$$= \lim_{h \rightarrow 0} \underbrace{g(x+h)}_{g(x)} \underbrace{\frac{f(x+h) - f(x)}{h}}_{f'(x)} + f(x) \underbrace{\frac{g(x+h) - g(x)}{h}}_{g'(x)}$$

$$= g(x)f'(x) + f(x)g'(x)$$

推论: ① 设 $c \neq 0$ 为常数, 则 $(cf(x))' = f'(x)$,

② 乘法法则也可推广至有限个函数的乘法.

$$\begin{aligned} \text{①} \quad (cf(x))' &= (c)' f(x) + c(f(x))' \\ &= cf'(x) \end{aligned}$$

$$\begin{aligned} \text{②.} \quad (f_1(x) f_2(x) f_3(x))' &= f_1'(x) f_2(x) f_3(x) + f_2'(x) f_1(x) f_3(x) \\ &\quad + f_3'(x) f_1(x) f_2(x), \end{aligned}$$

$$(f_1(x) \cdots f_n(x))' = \sum_{i=1}^n \left(\frac{f_1(x) \cdots f_n(x)}{f_i(x)} \right) f_i'(x)$$

$$f(x) = \underline{x^2 \cos x}$$

$$\begin{aligned} f'(x) &= (x^2)' \cos x + x^2 (\cos x)' \\ &= 2x \cos x - x^2 \sin x \end{aligned}$$

$$f(x) = x \sin x + 7\sqrt{x}$$

$$f'(x) = (x \sin x)' + (7\sqrt{x})'$$

$$\begin{aligned} &= (x)' \sin x + x (\sin x)' + 7(\sqrt{x})' \\ &= \sin x + x \cos x + \frac{7}{2\sqrt{x}} \end{aligned}$$

③ (除法法则) $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$

$$d\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)df(x) - f(x)dg(x)}{g^2(x)}$$

证明: $f(x)$ 和 $g(x)$ 在 x 处可导 $\Rightarrow$ 连续. $g(x) \neq 0$. $\lim_{h \rightarrow 0} g(x+h) = g(x) \neq 0$.

当 h 充分小时 $g(x+h) \neq 0$.

$$\begin{aligned} \left(\frac{f(x)}{g(x)}\right)' &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{f(x+h)g(x) - \overbrace{f(x)g(x+h)}}{g(x+h)g(x)} \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{g(x+h)g(x)} \cdot \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{g(x+h)g(x)} \left[g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h} \right] \\ &\quad \downarrow \\ &= \frac{1}{g^2(x)} \left[g(x) f'(x) - f(x) g'(x) \right] \end{aligned}$$

推论: $\left(\frac{1}{f(x)}\right)' = -\frac{f'(x)}{f^2(x)} \quad (f(x) \neq 0)$

$$\left(\frac{1}{f(x)}\right)' = \frac{(1)'f(x) - 1 \cdot (f(x))'}{f^2(x)} = -\frac{f'(x)}{f^2(x)}$$

$$\left(\frac{f(x)}{g(x)} \right)' = \left(f(x) \cdot \frac{1}{g(x)} \right)' = f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{g(x)} \right)'$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \left(-\frac{g'(x)}{g^2(x)} \right)$$

$$y = \tan x$$

$$(\tan x)' = \left(\frac{\sin x}{\cos x} \right)' = \frac{(\sin)' \cos x - \sin x (\cos)'}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \sec^2 x$$

$$(\cot x)' = -\frac{1}{\sin^2 x} = -\csc^2 x$$

$$f(x) = x^2 \sin x \ln x + \frac{\tan x}{x}$$

$$f'(x) = (x^2 \sin x \ln x)' + \left(\frac{\tan x}{x} \right)'$$

$$= (x^2)' \sin x \ln x + (\sin x)' x^2 \ln x$$

$$+ (\ln x)' x^2 \sin x + \frac{(\tan x)' x - \tan x (x)'}{x^2}$$

$$= 2x \sin x \ln x + \cos x x^2 \ln x$$

$$+ x \sin x + \frac{x \sec^2 x - \tan x}{x^2}$$

推论: ① 若 $f(x)$ 在 x 处可导, $g(x)$ 在 x 处不可导, 则 $f(x) \pm g(x)$ 在 x 处不可导.

② 设 $g(x)$ 在 x_0 处可导, $h(x)$ 在 x_0 处不可导, 则 $f(x) = g(x)h(x)$ 在 x_0 处可导的充分必要条件是 $g(x_0) = 0$, 从而有 $f'(x_0) = g'(x_0)h(x_0)$.

③ 若 $g(x)$ 在 x_0 处可导, $g(x_0) = 0$, $h(x)$ 在 x_0 处连续, 则 $f(x) = g(x)h(x)$ 在 x_0 处可导, 且 $f'(x_0) = g'(x_0)h(x_0)$.

④ 若 $g(x)$ 在 x_0 处可导, $g(x_0) = 0$, $g'(x_0) = 0$, $h(x)$ 在 x_0 附近有界, 则 $f(x) = g(x)h(x)$ 在 x_0 处可导, 且 $f'(x_0) = 0$.

证明: ② (如同乘法法则证明)

$$\underline{\underline{f'(x_0) = \lim_{t \rightarrow 0} \frac{g(x_0+t) - g(x_0)}{t} h(x_0+t) + g(x_0) \frac{h(x_0+t) - h(x_0)}{t}}}} \quad \text{③式}$$

由于 $\lim_{t \rightarrow 0} \frac{h(x_0+t) - h(x_0)}{t}$ 不存在

$$\Rightarrow f'(x_0) \text{ 存在} \Leftrightarrow g(x_0) = 0$$

④. 如同上式

证毕.

$$\text{由于 } g'(x_0) = 0 \Rightarrow \lim_{t \rightarrow 0} \frac{g(x_0+t) - g(x_0)}{t} = 0$$

由于 $h(x_0+t)$ 是关于 t 的局部有界函数

$$\Rightarrow \lim_{t \rightarrow 0} \frac{g(x_0+t) - g(x_0)}{t} h(x_0+t) = 0$$

$$\Rightarrow f'(x_0) = 0$$

例: 设 $f(x) = (x-a)\varphi(x)$, $\varphi(x)$ 在 $x=a$ 处连续, 求 $f'(a)$.

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x-a} = \lim_{x \rightarrow a} \frac{(x-a)\varphi(x)}{x-a} = \lim_{x \rightarrow a} \varphi(x) = \varphi(a)$$

$$(x-a) \quad \varphi(x)$$

$$a-a=0$$

$$(x-a)'|_{x=a} = 1$$

$$f'(x) = (x-a)'|_{x=a} \varphi(a)$$

$$= \varphi(a)$$

例: 设 $f(x) = \underbrace{(e^x-1)(e^x-2) \cdots (e^{nx}-n)}_{\text{product}}$, $n \in \mathbb{Z}^+$, 求 $f'(0)$.

$$g(x) = e^x - 1 \quad h(x) = (e^x - 2) \cdots (e^{nx} - n)$$

$$g(0) = 0.$$

$$g'(0) = e^x|_{x=0} = 1.$$

$$f'(0) = \underline{\underline{g'(0) h(0)}} = (1-2)(1-3) \cdots (1-n)$$

$$= (-1)(-2) \cdots (-(n-1))$$

$$= (-1)^{n-1} (n-1)!$$